

Extracting Economic Signals from ECB Communication with a Large Language Model

Measurement, Validation, and Classification Reliability

Francesca Condò

August 2026

Abstract

This project develops and evaluates a large-language-model-based measurement pipeline for extracting structured economic signals from 2,770 European Central Bank speeches delivered between 1999 and 2025. Using a predefined coding scheme, the model classifies four dimensions of substantive attention: inflation, real activity, financial stability, and uncertainty, as well as two directional outlook measures for inflation and growth. The central methodological question is how such generated classifications should be validated before being treated as economic variables.

Four sequential validation exercises show that model–human disagreement cannot be interpreted independently of the reliability of the underlying coding task. In an identical-text comparison, agreement between two independent human coders was 0.694, while model–human agreement was 0.889 and 0.778. Subsequent comparison of the human codebook with the production prompt revealed inconsistencies in the operational definitions. After the coding scheme was reconciled and refined, two coders independently classified fifty previously unseen speeches and achieved overall agreement of 0.810. Existing model classifications agree with the two coders at 0.737 and 0.743 in this sample, although these comparisons do not place humans and the model under a fully common specification because the production extraction predates the final codebook and prompt revision.

A dictionary benchmark performs substantially worse than the language model against the initial human coding reference, with overall agreement of 0.728 compared with 0.944, yet correlates much more strongly with contemporaneous inflation. This apparent reversal reflects differences in what the two measures capture, as the dictionary responds primarily to the level of inflation, whereas the LLM measure is designed to capture its expected trajectory. External validation against euro-area macroeconomic series produces relationships that are directionally consistent with the intended interpretation of the measures, but none of the five pre-specified associations survives adjustment for multiple testing. The results therefore support a measurement rather than a causal interpretation: LLM-derived classifications can provide useful economic signals, but their validity depends jointly on model performance, the reproducibility of the coding specification, and the construct that the measurement procedure is intended to recover.

Note on status. This is a completed methodological and technical project intended to develop and evaluate an LLM-based text-measurement pipeline, rather than a finished empirical research paper on ECB communication. The production classifications used throughout the analysis were generated under the original extraction specification and are intentionally retained as fixed inputs. The final validation round instead applies a subsequently revised human coding scheme that incorporates clarifications developed during the earlier validation exercises. Consequently, the round-four model–human comparisons should not be interpreted as estimates of relative reliability under a fully common specification. Re-extracting the validation sample, and potentially the full corpus, under the revised prompt would constitute a further extension rather than a prerequisite for the methodological results reported here.

Contents

Part I: Measurement and Validation	4
1 Introduction	4
1.1 Measurement Framework	4
1.2 Main Findings	5
2 Data and Sample Construction	5
2.1 The Speech Corpus	5
2.2 Sample Restrictions and Measurement Considerations	6
2.3 The Speech-Level Panel	6
3 The Measurement Instrument	7
3.1 Coding Scheme	7
3.2 LLM Extraction	7
4 Internal Validation	7
4.1 Round One: Initial Non-Blind Audit	8
4.2 Round Two: Blind Full-Text Validation	9
4.3 Round Three: Independent Second Coding	10
4.4 Reconciliation of the Coding Specifications	11
4.5 Round Four: Reproducibility Under the Revised Coding Scheme	12
4.6 Model-Human Agreement Under the Revised Coding Scheme	14
4.6.1 Truncation and Statistical Uncertainty	14
4.7 Limitations of the Validation Evidence	15
5 Misclassification Correction	15
5.1 Correction Framework	16
5.2 Application of the Misclassification Correction	16
5.3 Interpretation and Limitations	17
6 From Speech-Level Classifications to Communication Measures	18
6.1 Construction of the Measures	18
6.2 Dispersion and Polarization	19
6.3 Summary of the Communication Measures	19
6.4 Distinct Information in Attention and Outlook	19
7 External Validation Against Euro-Area Macroeconomic Data	21
7.1 Validation Strategy	21
7.2 External Validation Data	21
7.3 Statistical Inference	22
7.4 Pre-Specified Validation Relationships	23
7.5 Inflation Attention and Distance from the Reference Level	23
7.6 Growth Outlook and Unemployment	25

7.7	Robustness to Speech Length	25
7.8	Interpretation and Limitations	26
8	Benchmarking Against a Keyword Dictionary	26
8.1	Benchmarking Strategy	26
8.2	Construction of Dictionary Benchmark	27
8.3	Agreement with Human Coding	27
8.4	External Comparison of the Outlook Measures	28
8.5	Comparison between the Contemporaneous Correlations	29
8.6	Interpretation and Limitations	29
9	Conclusion	30
	 Part II: Statistical and Methodological Appendix	 33
A	Validation Statistics and Misclassification	33
A.1	Agreement and Classification Rates	33
A.2	Recovery Under Choice-Based Validation Sampling	34
A.3	Exact Intervals for Validation Proportions	35
A.4	Misclassification Correction	35
A.5	Uncertainty in Corrected Shares	36
A.6	Dependence Between Attention and Outlook Classifications	37
B	HAC Bandwidth and Overlapping Outcomes	37
C	Dictionary Benchmark Construction	38
C.1	Attention as a Normalized Density	38
C.2	Direction within Topic Sentences	38
C.3	Prevalence-Matched Thresholds	39
C.4	Neutral-Band Calibration	39
	 Part III: Measurement and Replication Materials	 40
D	Reproducing the Results	40
E	Production LLM Extraction Prompt	41
F	Revised LLM Extraction Prompt	42
G	Codebook Revision	45

Part I: Measurement and Validation

1 Introduction

Economic text contains information that is not directly available in conventional structured datasets, motivating a growing literature on methods for converting text into quantitative variables (Gentzkow et al., 2019). Recent advances in large language models (LLMs) extend this approach by allowing contextual classification and annotation of text across a wide range of social-science tasks (Ziems et al., 2024). Central-bank communication provides a natural application for such methods because policy communication can convey information about both monetary policy and the central bank’s assessment of economic conditions. A substantial literature studies the economic role of central-bank communication (Blinder et al., 2008), while text-based approaches have shown that different dimensions of communication can contain distinct information (Hansen and McMahon, 2016).

This project explores the use of LLMs as measurement instruments for economic text analysis. Specifically, it develops and evaluates an LLM-based classification pipeline for extracting structured measures from European Central Bank speeches. The objective is not to estimate the effects of ECB communication or to establish substantive conclusions about it, but to examine how such dimensions of communication can be constructed and validated as quantitative economic measures. The project is therefore intended as a methodological and technical exercise in constructing, validating, and analyzing LLM-derived economic variables.

The central methodological issue is that an LLM-generated classification is a constructed measure whose reliability depends on the ability of the classifier to recover the information that a careful human reader would identify in the same text, different from directly observed economic data. Thus, before such classifications can be used in downstream economic analysis, their measurement properties need to be assessed. This is particularly important because the performance of LLM-based annotation varies across tasks and evaluation settings (Ziems et al., 2024). The primary purpose of the project is therefore to demonstrate the construction and evaluation of an LLM-based text-measurement pipeline and to make explicit the steps required before its outputs can be treated as economic variables.

The analysis focuses on three aspects of the measurement procedure: the extent to which LLM classifications reproduce human coding of the same speeches, the nature and direction of classification errors when model and human assessments diverge, and the performance of the resulting measures relative to simpler and less costly text-classification approaches. The project addresses these questions through an iterative validation procedure combining LLM extraction, independent human coding, inter-coder agreement analysis, error diagnostics, and comparison with a dictionary-based benchmark. External economic series are additionally used to examine whether the constructed measures display relationships consistent with their intended interpretation.

1.1 Measurement Framework

The main dataset is the European Central Bank’s official precompiled speech dataset, covering 2,770 speeches delivered between January 1999 and December 2025. Six signals were extracted from each speech under a fixed coding scheme: binary attention indicators for inflation, real activity, financial stability and uncertainty, together with three-class directional outlook variables for inflation and growth taking values in $\{-1, 0, +1\}$. The extraction used `gpt-5-mini` under the established prompt (Appendix Section E) and is treated throughout as a fixed input, as it is never regenerated. A revised version is proposed in Appendix Section F

The extraction is part of a broader measurement and validation framework. A predefined coding scheme establishes the interpretation of each variable and the criteria required for classification. Human-coded validation samples are used to assess agreement between the LLM classifications and independent human annotations and to characterize observed classification errors. The speech-level classifications are then aggregated into quarterly communication measures, keeping the intensity of attention, the volume of communication, and the conditional direction of the outlook conceptually distinct, without collapsing them into a single measure. External macroeconomic series are used to assess the relationship between the constructed measures and relevant economic outcomes, while a dictionary-based benchmark

provides a comparison with a simpler and less computationally intensive text-classification approach.

1.2 Main Findings

The analysis provides evidence on the measurement properties of the LLM-based classification procedure. Human validation indicates that classification performance varies substantially across variables and, more importantly, that model–human disagreement cannot automatically be interpreted as model error. The introduction of a second independent coder revealed substantial disagreement between human classifications themselves, showing that validation against a single human reference can confound model error with ambiguity in the coding task. Comparison of the human codebook with the production prompt subsequently identified several discrepancies and decision boundaries requiring clarification. After the coding scheme was revised, the same two coders independently classified fifty previously unseen speeches and achieved an overall agreement rate of 0.810, compared with 0.694 on the round-three identical-text sample. Because the samples and sampling designs differ, this increase cannot be attributed exclusively to the revised rules, but it provides descriptive evidence of greater reproducibility under the revised coding procedure.

Comparison with a dictionary-based approach also shows meaningful differences between the two classification methods, particularly for variables whose identification depends on contextual interpretation rather than the presence of individual terms. Against the initial human reference, model–human agreement is 0.944, compared with 0.728 for the dictionary, although the relative advantage varies substantially across variables. Inflation attention is classified equally well by both methods, while larger differences emerge for growth attention and uncertainty. The inflation-outlook measures also appear to capture different information: the dictionary is substantially more strongly associated with the contemporaneous HICP level, at 0.618 compared with 0.229 for the model, while the model is relatively more associated with recent directional changes in inflation. The resulting measures should therefore not be regarded as interchangeable even when they are intended to represent the same broad underlying concept, and a stronger correlation with an external economic series should not by itself be interpreted as evidence of a better measure.

The external-validation results are more limited but broadly consistent with the intended economic interpretation of the constructed measures. Attention and conditional outlook behave as empirically distinct dimensions of communication, and all five pre-specified relationships with external macroeconomic variables have the expected direction. None of the individual associations, however, survives adjustment for multiple testing. These results are therefore treated as evidence on the economic coherence of the measurement framework rather than as established empirical relationships. In particular, the analysis does not identify causal effects of ECB communication or establish forecasting performance.

The project is intended as a methodological and technical exercise rather than a finished empirical study, and several limitations remain deliberately unresolved. Most importantly, the production extraction was generated before the final revision of the coding scheme and has not been repeated under the complete revised specification. The misclassification correction is consequently retained as an illustration rather than as a preferred corrected series, and uncertainty from the extraction stage is not propagated into downstream inference. A subsequent iteration would be able to re-extract the round-four speeches under the revised prompt, compare model and human classifications under a common specification, and use that validation to determine whether a full-corpus re-extraction and updated measurement-error correction are warranted.

2 Data and Sample Construction

2.1 The Speech Corpus

As noted above, the speech data come from the European Central Bank’s official precompiled speeches dataset, downloaded programmatically by `src/download_data.py`. The raw file is not committed to the repository; the pipeline regenerates it. After cleaning, the corpus contains 2,770 speeches spanning 14 January 1999 to 19 December 2025, delivered by 26 distinct primary speakers across four presidencies.

Table 1 shows how the corpus distributes across presidential terms, together with the mean speech

length. The distribution is uneven in a way that matters for anything said about trends: the Trichet and Draghi presidencies together account for two thirds of the corpus, and mean speech length falls markedly across the sample.

Table 1: Corpus composition by presidency

Presidency	Speeches	Mean words
Willem F. Duisenberg (1998–2003)	373	3,628
Jean-Claude Trichet (2003–2011)	947	3,291
Mario Draghi (2011–2019)	940	2,694
Christine Lagarde (2019–)	510	2,957
Total	2,770	3,072

Macroeconomic series used for external validation are obtained from the ECB Data Portal through its SDMX REST API. The analysis uses the euro-area HICP annual inflation rate, the unemployment rate, and the ECB deposit facility rate. Monthly HICP inflation and unemployment observations are averaged within each quarter, while the deposit facility rate is forward-filled between policy changes and measured at the end of each quarter. The main refinancing operations rate is additionally constructed as a robustness measure.

2.2 Sample Restrictions and Measurement Considerations

Three characteristics of the speeches dataset require additional treatment in the construction and interpretation of the measures: short items, variation in speech length, and language.

Short items. Fifty-three documents contain fewer than 300 words. Inspection indicates that these primarily consist of slide decks, press-briefing materials, and agenda items rather than substantive speeches. Because their inclusion could affect aggregate attention measures while contributing little comparable textual content, they are excluded from the baseline sample, leaving 2,717 speeches. The exclusion is implemented through a sample flag rather than by removing the observations, allowing the full sample to be reconstructed and the restriction to be varied in robustness analysis.

Speech length and truncation. The length of the speeches supplied to the classification model is capped at 2,000 words, a limit that binds for 70.3% of speeches. Speech length varies substantially over the sample period: mean length declines from approximately 4,062 words in 1999 to 2,460 words in 2017, a reduction of about 39%. This variation is potentially relevant for the attention measures because longer texts provide more opportunities for a topic to receive substantive discussion. Changes in measured attention over time could therefore partly reflect changes in speech length rather than changes in communication content. Trend-based interpretations of the attention measures consequently require consideration of speech length, and Section 7.7 examines this issue through an explicit length adjustment.

Language. A rule-based language check (function-word heuristic) based on the prevalence of common English function words identifies 151 speeches as potentially non-English, concentrated primarily in the earlier part of the sample. Differences in language could affect the consistency of the classification procedure and therefore constitute a potential source of measurement variation. As a robustness check, the quarterly measures are reconstructed using only speeches classified as English. These series correlate between 0.98 and 0.996 with their corresponding baseline measures. The English-only sample is therefore retained as a robustness specification, while the full sample remains the baseline, given the high correlation between the two.

2.3 The Speech-Level Panel

The speech-level panel is constructed by merging the extracted signals with the corresponding speech metadata (`analysis/10_panel.py`). Additional variables are derived to capture characteristics relevant to the measurement and subsequent analysis, including speech length and its logarithm, an indicator for

whether the extraction limit binds, the language flag, the number of speakers, and indicators for speeches delivered by the ECB President or Chief Economist in office at the time of the speech. Office assignments are constructed from date-specific lookup tables and validated against the speaker information in the underlying metadata.

3 The Measurement Instrument

3.1 Coding Scheme

The coding scheme in `docs/coding_rules.md` was written before extraction and defines six variables. Four are binary attention indicators, taking the value one when the topic is a fundamental part of the discussion and zero otherwise, for inflation and price developments, real economic activity, financial stability, and uncertainty. Two are three-class outlook variables for inflation and growth, taking -1 for a downside or weakening emphasis, $+1$ for an upside or strengthening emphasis, and 0 for balanced, mixed, purely descriptive or absent discussion.

Three features of the scheme are to be analyzed as they have important consequences in the subsequent analysis. First, in depth discussion is required rather than keyword presence, and passing references are explicitly insufficient. Second, the outlook variables capture the expected direction of future developments rather than current economic conditions: the scheme instructs the coder to classify the direction of the expected path “not merely whether the current level of inflation is high or low”. The classification therefore reflects the dominant forward-looking trajectory of inflation or economic activity, rather than whether their current levels are high or low. Third, the attention and outlook variables are subject to a logical constraint: when attention to a topic is zero, the corresponding outlook is necessarily coded as zero. As a result, errors in the two measures are not independent. A disagreement in the attention classification can mechanically generate a disagreement in the corresponding outlook classification, a feature that is explicitly considered in the validation analysis.

The classification is additionally restricted to information contained in the supplied text, excluding outside knowledge about the speaker, the date, economic conditions, or subsequent events. This restriction is important for external validation. If the model were to use prior knowledge of contemporaneous macroeconomic conditions when assigning classifications, the resulting measures could correlate with external economic series independently of the information contained in the speech itself. Such contamination would undermine the interpretation of correlations between the constructed indices and economic variables as evidence of the validity of the text-based measures.

3.2 LLM Extraction

Extraction is performed using `gpt-5-mini` under the classification prompt reported in Appendix Section E, with model outputs constrained by a predefined schema, ensuring that the returned output is machine-readable and that each variable takes an admissible value. The schema therefore addresses output consistency and a class of potential implementation errors, while the validity of the resulting measures must be assessed separately through the validation exercises described below.

The extracted classifications are treated as fixed inputs for subsequent analysis: the baseline extraction is not regenerated across alternative prompt specifications, although a revised version is written at the end, implementing possible improvements given the results of the analysis (Appendix Section F). The absence of a systematic prompt-sensitivity analysis therefore remains a limitation of the current exercise, which is not carried out primarily because repeated extraction of the full corpus would impose additional computational costs and time requirements.

4 Internal Validation

The internal validation proceeds through four sequential rounds. All four stages are documented, not because each provides equally informative evidence in isolation, but because their progression and interaction provide insight into the development and refinement of the extraction procedure. This is

consistent with the purpose of the project as a technical and methodological exercise rather than a completed empirical study.

Table 2: Design of the four internal-validation rounds

Round	N	Human coders	Blind?	Human text	Sampling	Rules
1. Initial non-blind audit	30	A	No	Same 2,000-word excerpt as the model	Non-seeded audit sample	Original
2. Blind full-text validation	60	A	Yes	Complete speech	Seeded; stratified by period and model-positive attention count	Original
3. Independent second coding	53	A and B	B: yes	A: complete speech; B: same 2,000-word excerpt as the model	Subsample of round two	Original
4. Revised-scheme validation	50	A and B	Yes	Identical complete speech for both coders	New sample; flat sampling weights	Revised

Notes: Round three reuses coder A’s full-text classifications from round two and adds independent coding by coder B using the text supplied to the model. For 12 of the 53 speeches, the 2,000-word extraction limit does not bind, so the model and both human coders observe identical text. Round four uses 50 speeches not included in any preceding validation round. The existing model classifications used in the round-four comparison were generated under the production prompt and therefore do not implement the complete revised specification used by the human coders.

4.1 Round One: Initial Non-Blind Audit

Thirty speeches, spanning December 2000 to September 2025 and covering sixteen speakers, were independently coded by hand according to the predefined coding scheme. For each variable, the human classification was recorded alongside an assessment of the model–human comparison: **agree** when the classifications coincided, **error** when they differed and the model classification was judged incorrect, and **borderline** when the disagreement reflected genuine ambiguity in the coding rules. Before computing validation statistics, the scoring procedure performs an internal consistency check between the recorded classifications and agreement flags, preventing transcription inconsistencies from entering the analysis.

Overall cell-level agreement was $170/180 = 0.944$, with Cohen’s κ (Cohen, 1960) ranging from 0.773 to 1.000 across variables. Estimated sensitivity was 1.000 for each attention measure, while specificity was 0.850 for inflation attention, 0.857 for growth attention, and 1.000 for financial-stability and uncertainty attention. All five disagreements in the outlook variables involved a directional model classification and a neutral human classification; no disagreement involved a reversal in direction.

Several limitations restrict the interpretation of these results. The coding was not blind, relied on a single human coder, and was based on a sample of only thirty speeches that had not been selected through a reproducible seeded sampling procedure. The resulting agreement statistics therefore provide limited evidence about classification performance outside this initial sample.

A further limitation concerns the text presented to the human coder. The coder evaluated the same 2,000-word excerpts supplied to the model, and 25 of the 30 sampled speeches exceeded this limit. Consequently, the exercise could assess agreement between the model and the human coder conditional on the information contained in the extraction excerpt, but could not detect classification errors caused by truncation. If relevant discussion occurred beyond the 2,000-word limit, both the model and the human coder could assign the same classification despite that classification not representing the complete speech. The estimated sensitivity of 1.000 should therefore not be interpreted as evidence of perfect sensitivity with respect to the full speech text.

The sample also contained one 69-word item that falls outside the baseline sample and contributes six relatively uninformative agreements. Excluding this observation reduces overall cell-level agreement

slightly, from 0.944 to 0.9425.

Given these limitations, the first validation round is treated as a development and diagnostic exercise rather than as the primary evidence on classification accuracy. Its findings motivated the subsequent validation rounds, which introduce reproducible sampling, full-text human coding, and additional independent coding.

4.2 Round Two: Blind Full-Text Validation

The second validation round uses a reproducibly selected sample of sixty speeches. Sampling is stratified by period and by the number of attention variables assigned a positive classification by the model, with greater representation of speeches receiving multiple positive attention classifications. Human coding is conducted using the complete speech text, and the coding sheet contains no information about the corresponding model classifications.

This sampling design differs from that used in the first round and must be taken into account when interpreting the resulting validation statistics. Because selection is related to the model classifications, unweighted sample agreement and conventional sample estimates of sensitivity and specificity need not represent their population counterparts. For a binary classification, the relevant population quantities are related through

$$p = \text{PPV} \cdot q + (1 - \text{NPV})(1 - q), \quad (1)$$

$$\alpha = \text{PPV} \cdot q/p, \quad \beta = \text{NPV} \cdot (1 - q)/(1 - p). \quad (2)$$

where $q = P(M = 1)$ is the population prevalence of positive model classifications, $p = P(H = 1)$ is the implied prevalence under the human classification, and α and β denote sensitivity and specificity, respectively. The model-positive rate q is observed directly in the full baseline sample. Intuitively, Positive Predictive Value (PPV) and Negative Predictive Value (NPV) describe how often the human coder agrees with the model conditional on the model’s classification; combining these quantities with the population frequency of model-positive classifications allows the same joint distribution to be expressed in terms of sensitivity and specificity, which instead condition on the human classification. The interpretation of the remaining quantities, however, depends on the stratified sampling design and is revisited in the subsequent validation rounds. For the same reason, raw agreement in this sample is not directly comparable with the 0.944 agreement reported in round one.

Table 3: Round two: rates recovered from predictive values

Variable	PPV	NPV	Sensitivity	Specificity
Inflation attention	0.737	1.000	1.000	0.718
Growth attention	0.714	1.000	1.000	0.446
Financial stability	0.696	1.000	1.000	0.569
Uncertainty	0.487	1.000	1.000	0.606

Two features of the second round are particularly informative for the subsequent validation exercise.

No false negatives are observed in the validation sample. Across the sixty speeches, no human-positive attention classification occurs among the observations classified as negative by the model. This result is obtained from full-speech human coding and therefore does not share the truncation problem of the first validation round. Forty-one of the sixty speeches exceed the 2,000-word model-input limit, with a median of 1,194 words omitted among truncated speeches. The result therefore provides evidence that truncation does not generate false negatives within this sample. It should not, however, be interpreted as establishing population sensitivity of exactly 1.000.

Sign reversals occur in the outlook classifications. In contrast with the first round, the second validation sample contains one inflation-outlook sign reversal as well as three cases in which the human coder assigns a directional growth outlook where the model assigns a different classification. Most outlook disagreements nevertheless involve differences between a directional and a neutral classification: six of seven inflation-outlook disagreements and sixteen of nineteen growth-outlook disagreements take this form.

The logical dependence between attention and outlook classifications is also visible in these disagreements. One of the seven inflation-outlook disagreements and six of the nineteen growth-outlook disagreements arise from a disagreement in the corresponding attention classification. Because the coding scheme requires the outlook to equal zero whenever attention equals zero, these observations do not constitute independent evidence about directional outlook classification.

Taken in isolation, the second-round results appear to indicate substantially lower specificity than the initial audit. Subsequent validation, however, shows that this interpretation is incomplete: disagreement between the model and a single human coder does not by itself establish model error, particularly where the underlying coding rules permit different reasonable interpretations. This motivates the additional independent coding and adjudication introduced in the following rounds.

4.3 Round Three: Independent Second Coding

A second independent coder evaluated fifty-three of the speeches included in the previous validation sample, using the coding rules in `docs/coding_rules.md`. The coder evaluated the same 2,000-word excerpts supplied to the model. For twelve of these speeches, the extraction limit did not bind, so the model and both human coders evaluated the complete and identical text.

Table 4: Positive classification rates by coder, fifty-three shared speeches

Variable	Coder A	Coder B	Model
Inflation attention	0.491	0.736	0.660
Growth attention	0.585	0.868	0.830
Financial stability	0.509	0.868	0.755
Uncertainty	0.321	0.868	0.660
Inflation outlook	0.170	0.132	0.283
Growth outlook	0.321	0.491	0.547

The first four rows report the share of speeches assigned a positive attention classification. For the ternary outlook variables, the final two rows report the share assigned either directional classification rather than zero. For forty-one of the fifty-three speeches, coder B evaluated the 2,000-word model input while coder A evaluated the complete speech. Differences between the two coders in this sample may therefore reflect both differences in coding behavior and differences in the text observed.

Table 5: Agreement on identical text, twelve speeches, seventy-two cells

Pair	Agreement	95% interval
Human A vs Human B	0.694	[0.575, 0.798]
Model vs Human A	0.889	[0.793, 0.951]
Model vs Human B	0.778	[0.664, 0.867]

Because the model and second coder both read the same excerpt, model-versus-B can additionally be computed on all fifty-three speeches, where it pools to 0.739.

Model-human agreement is at least as high as the observed agreement between the two human coders in this sample. This should not be interpreted as evidence that the model is more accurate than either human coder, since human-human agreement provides a reference point for coding consistency rather than a ground-truth measure of classification accuracy. The two human coders differ substantially in their propensity to assign positive classifications: for three of the four attention variables, coder B assigns a positive classification to approximately 87% of speeches, compared with rates between 32% and 59% for coder A. The observed agreement rates therefore partly reflect systematic differences in coding tendencies across human annotators. The more limited conclusion is that model-human agreement is comparable to the level of agreement observed between the two human coders.

The aggregate agreement rates, however, conceal substantial variation across individual variables. While Table 5 reports raw agreement, the variable-level comparison below uses Cohen’s κ to account for agreement expected from the marginal distribution of classifications (Cohen, 1960). The speeches to which the table refers are the twelve untruncated speeches for which the model and both human coders observed identical text.

Table 6: Human–human and model–human agreement by variable (Cohen’s κ)

Variable	Human A–B	Model–A
Financial stability	−0.154	1.000
Inflation outlook	0.059	0.789
Uncertainty	0.182	0.824
Growth attention	0.500	0.500
Inflation attention	0.636	0.824
Growth outlook	0.714	0.714

Table 6 shows substantial heterogeneity in agreement across variables. Human–human agreement, measured by Cohen’s κ , ranges from −0.154 for financial stability to 0.714 for the growth outlook, indicating that some variables are considerably more consistently interpreted by the two human coders than others. Model–human agreement is equal to or higher than human–human agreement for every variable in this sample, ranging from 0.500 to 1.000.

Human–human agreement is particularly low for financial stability, with a negative Cohen’s κ , and remains weak for inflation outlook and uncertainty, indicating that classifications for these variables are sensitive to differences in interpretation. By contrast, growth outlook exhibits relatively high and identical agreement in both comparisons. The table therefore provides evidence that the reliability of the coding task varies across variables and that model performance should be evaluated relative to the level of agreement observed between independent human coders.

The principal implication of this round concerns the interpretation of model–human disagreement in the preceding validation exercise. Because independent human coders themselves disagree on a non-negligible share of classifications, disagreement between the model and a single human coder cannot in general be interpreted as model error without further adjudication. This motivates the fourth validation round.

The results of Round 3 change the interpretation of the error rates estimated in Round 2. Because the human reference classification is itself subject to inter-coder disagreement, those estimates measure model performance relative to coder A rather than classification error against an unambiguous ground truth. More generally, validation against a single human coder cannot distinguish model error from ambiguity in the coding task. Human–human agreement therefore provides an important benchmark for interpreting model–human agreement and, where disagreement is substantial, motivates clarification of the coding rules before model errors are estimated or corrected.

4.4 Reconciliation of the Coding Specifications

The disagreement observed in round three motivated a direct comparison between the human coding rules in `docs/coding_rules.md` and the production prompt in `prompts/extraction_prompt.txt`. The two specifications were substantively similar, but the comparison identified several differences in the treatment of financial stability and uncertainty that could affect classification.

The clearest discrepancy concerned an exclusion rule for technical, operational, legal, credit, settlement, and financial-system risks. In the production prompt, this rule belongs to the definition of uncertainty attention: the presence of such risks is not sufficient for an uncertainty classification unless the text also emphasizes unpredictability, unusually wide possible outcomes, or difficulty assessing the economic outlook. In the human codebook, the clause had been placed under financial stability. In that position, it conflicted with the financial-stability definition, which explicitly treats systemic risk, financial fragility, and related vulnerabilities as qualifying evidence. Its displacement also left the uncertainty definition without the corresponding exclusion.

A second, more limited difference concerned references to financial markets, interest rates, and monetary-policy transmission. Both specifications exclude ordinary discussion of these topics from financial-stability attention. The production prompt, however, contains an explicit exception when systemic or stability-related concerns are substantively addressed. This qualification was absent from the human codebook. Taken literally, the human specification could therefore assign zero to cases in which financial markets or transmission provide the context for a substantive discussion of systemic risk, while the production prompt permits a positive classification.

These differences provide a plausible explanation for part of the disagreement observed in the preceding validation rounds. In the round-three sample, coder A assigns financial-stability attention to 50.9% of speeches, compared with 75.5% for the model, while coder B assigns it to 86.8%. Substantial differences are also observed for uncertainty attention. These patterns are consistent with the possibility that small differences in the operational definitions generated different classification thresholds across coders.

The comparison therefore led to a reconciliation and refinement of the coding specifications rather than a change in the underlying constructs. The revised codebook (Appendix Section G) restores the misplaced uncertainty exclusion, incorporates the missing financial-stability qualification, and adds decision rules for ambiguities identified during the validation process. Round four evaluates whether these revisions improve the consistency with which independent coders apply the classification scheme.

4.5 Round Four: Reproducibility Under the Revised Coding Scheme

Table 7: Round-four agreement, fifty speeches, three hundred cells

Pair	Agreement	95% interval
Human A vs Human B	0.810	[0.761, 0.853]
Model vs Human A	0.737	[0.684, 0.786]
Model vs Human B	0.743	[0.691, 0.792]

Round four examines the reproducibility of classification under the revised coding scheme. The six additional decision rules were finalized following the analysis of round-three disagreements and before the round-four sample was drawn. The revised specification was then evaluated on a new set of fifty speeches that had not appeared in any of the preceding validation rounds. The same two coders independently classified each speech, both working blind and from identical full text.

The sampling design also differs from round two. The earlier validation deliberately oversampled speeches receiving positive model classifications because its objective was to investigate model classification errors. Round four instead focuses on the coding task itself: whether two independent coders applying the same revised rules to the same text produce consistent classifications. Sampling weights are therefore flat rather than determined by the model’s classifications. The realized sample nevertheless reproduces several corpus-level model classification rates closely. For example, the model-positive rate for inflation attention is 0.600 in both the round-four sample and the baseline corpus, while the corresponding rates for uncertainty attention are 0.560 and 0.556.

Table 8: Human–human agreement under the round-three and round-four coding schemes

Variable	κ round 3	κ round 4	Difference
Financial stability	−0.154	0.560	+0.714
Inflation outlook	0.059	0.631	+0.572
Inflation attention	0.636	0.800	+0.164
Growth attention	0.500	0.662	+0.162
Uncertainty	0.182	0.280	+0.098
Growth outlook	0.714	0.650	−0.064

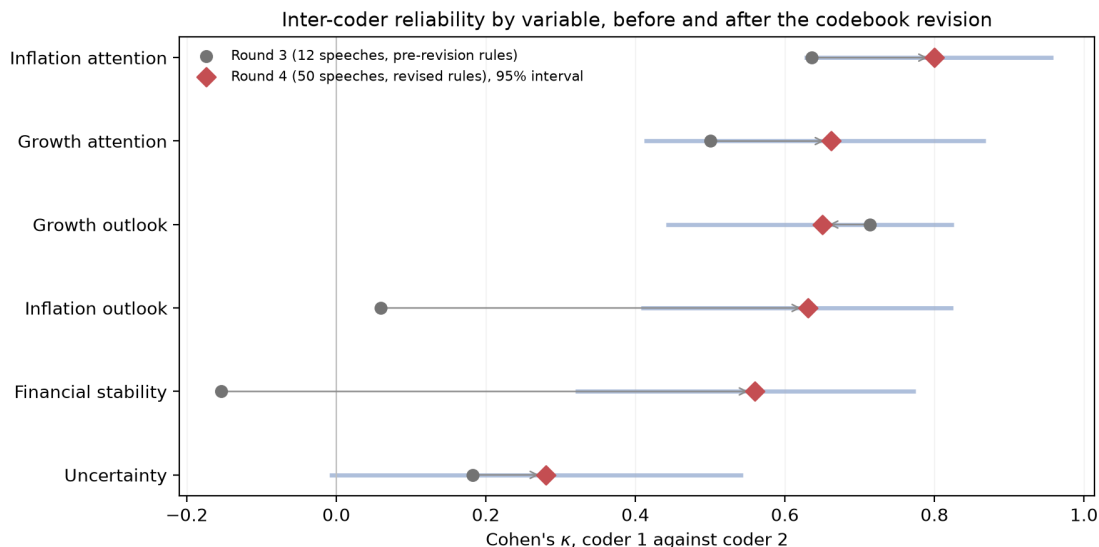


Figure 1: Inter-coder reliability by variable before and after the codebook revision.

Notes: The figure reports Cohen’s κ by variable for coder A and coder B. Grey circles report round-three estimates based on 12 speeches coded on identical text under the pre-revision rules. Red diamonds report round-four estimates based on 50 speeches coded on identical full text under the revised rules. Horizontal bars report 95% percentile-bootstrap confidence intervals obtained by resampling speeches. Arrows connect the round-three and round-four point estimates for each variable. Comparisons across rounds are descriptive because the samples, sampling designs, and coding specifications differ.

Across the six variables, the two coders agree on 243 of 300 round-four classifications, corresponding to an overall agreement rate of 0.810 with a 95% interval of [0.761, 0.853]. Human–human agreement is therefore higher than the 0.694 observed in the twelve-speech identical-text subsample from round three. Within each round, both coders apply a common specification to identical text, so the observed disagreement cannot be attributed to differences in the information available to them.

The difference between the two rounds should nevertheless be interpreted descriptively rather than as a causal estimate of the effect of revising the codebook. Round three and round four contain different speeches and were generated under different sampling designs in addition to using different coding specifications. As a check on the role of sample composition, restricting the round-four sample to speeches receiving positive model classifications on three or more topics—the region in which the round-three sample was concentrated—produces a similar human–human agreement rate of 0.817. The higher agreement observed in round four is therefore not explained solely by the broader composition of its sample, although the contribution of the revised rules cannot be separately identified.

At the variable level, inter-coder reliability is higher in round four for five of the six measures. The largest differences occur for financial stability, where Cohen’s κ increases from -0.154 to 0.560 , and inflation outlook, where it increases from 0.059 to 0.631 . These patterns are consistent with the revisions addressing ambiguities identified after round three, including the reconciliation of the financial-stability definition and the introduction of a rule for conflicting inflation horizons. Inflation and growth attention also exhibit higher agreement under the revised substantive-attention criteria. Growth outlook is the only variable for which κ is lower, declining from 0.714 to 0.650 .

Uncertainty remains the least reliable component of the coding scheme. Its κ increases from 0.182 to 0.280 , with a 95% interval of $[-0.009, 0.544]$, while raw agreement in round four is 0.700 . The targeted clarification of the uncertainty definition therefore does not coincide with the improvement observed for several other variables. This suggests that the remaining difficulty may extend beyond an individual decision rule and motivates further examination of whether the current definition combines conceptually distinct dimensions of uncertainty.

Overall, round four provides descriptive evidence of greater inter-coder reproducibility under the

revised coding procedure. The comparison is not a controlled test of the codebook revision, but the use of the same two independent coders, identical information within each round, a previously unseen validation sample, and the restricted-sample robustness check make a purely compositional explanation for the observed increase less plausible.

4.6 Model–Human Agreement Under the Revised Coding Scheme

The round-four sample also permits a comparison between the existing model classifications and human classifications produced under the revised coding scheme. Human–human agreement is 0.810, while model–human agreement is 0.737 against coder A and 0.743 against coder B. Thus, within the round-four sample, agreement between the two human coders is higher than agreement between the model and either human coder.

This ordering differs from the round-three identical-text comparison, where agreement was 0.694 between the two human coders, 0.889 between the model and coder A, and 0.778 between the model and coder B. However, the model–human agreement rates from the two rounds are not directly comparable. The round three three-way comparison is based on twelve untruncated speeches, corresponding to 72 speech–variable classifications, whereas round four contains fifty previously unseen speeches and 300 classifications. The samples were also generated under different sampling designs. The difference between the round-three and round-four model–human agreement rates therefore cannot be interpreted as a change in model performance.

The coding specifications also differ, although less substantially than in the earlier validation rounds. Several round-four revisions explicitly reconcile the human codebook with rules already contained in the production prompt, bringing the human coding procedure into closer alignment with the specification originally supplied to the model. The revised codebook also introduces six additional decision rules motivated by ambiguities identified during round three. These rules were finalized after the production extraction and were therefore not explicitly available to the model. The existing model outputs and the round-four human classifications consequently reflect closely related, but not fully identical, coding specifications.

The human–human comparison has a different interpretation. Within each round, the same two coders apply the same coding rules to identical text, making the comparison informative about the reproducibility of the coding procedure under the specification used in that round. As discussed above, the higher agreement observed in round four is consistent with improved reproducibility under the revised scheme, although differences in the underlying samples prevent the change from being attributed exclusively to the codebook revision.

The lower model–human agreement relative to human–human agreement within round four should therefore not be interpreted as evidence that the model is less reliable than human coding under a common specification. A direct comparison requires the model to classify the round-four speeches under the complete revised prompt, so that the model and both human coders apply the same rules to the same observations. Because this re-extraction has not yet been conducted, it remains a subsequent step of the project.

4.6.1 Truncation and Statistical Uncertainty

Round three also provides evidence on the effect of the 2,000-word extraction limit. For truncated speeches, disagreement between coders also reflects differences in the text available to them, as for untruncated speeches both coders observe identical text and disagreement reflects only differences in classification. Under the assumption that inter-coder disagreement is comparable across the two groups, the difference in disagreement rates provides an indication of the additional contribution of truncation. Averaged across variables, disagreement on identical text is 0.306, while the additional disagreement associated with truncated observations is 0.089, suggesting that differences in coding account for a larger share of observed disagreement than the information omitted by the 2,000-word limit.

Statistical uncertainty in the validation rates is reported using exact Clopper–Pearson intervals (Clopper and Pearson, 1934) rather than nonparametric bootstrap intervals. This distinction is particularly important for estimates at the boundary. When a validation sample contains no observed failures, a nonparametric bootstrap resamples only successes and therefore returns the degenerate interval

[1, 1], incorrectly suggesting that the underlying rate is known with certainty. Exact Clopper–Pearson intervals instead account for finite-sample uncertainty: for example, ten successes out of ten produce a point estimate of 1.000 but remain consistent, at the 95% confidence level, with a true success probability as low as approximately 0.69¹. The exact intervals therefore preserve the uncertainty that remains when perfect performance is observed in a small sample and prevent a sample estimate of 1.000 from being interpreted as evidence of perfect population performance. Confidence intervals for Cohen’s κ are treated separately and obtained using a percentile bootstrap over speeches, since κ is not a simple proportion and does not admit the exact binomial intervals used for the agreement rates.

4.7 Limitations of the Validation Evidence

The revised validation exercise increases the human-coded sample from the twelve identical-text speeches available in round three to fifty independently double-coded speeches in round four. This provides a substantially stronger basis for assessing inter-coder reproducibility, but several limitations remain. First, the estimate is based on a single pair of human coders. The observed agreement rate of 0.810 may therefore reflect characteristics of these particular coders as well as the underlying difficulty of the classification task. Additional independent coders would be required to determine how representative this level of agreement is across human annotators. Second, although pooled agreement is estimated with a relatively narrow 95% interval of [0.761, 0.853], uncertainty remains considerably greater at the variable level. For example, the confidence interval for Cohen’s κ is approximately [0.41, 0.87] for growth attention and [−0.01, 0.54] for uncertainty. Differences across individual variables should therefore be interpreted cautiously.

A further limitation concerns the comparison between human and model classifications. The round-four human coders apply the revised coding scheme, whereas the existing model classifications were generated before the six additional decision rules were introduced. Although part of the codebook revision brought the human specification into closer alignment with the production prompt, the two classification procedures are not fully identical. The round-four model–human agreement rates should therefore not be interpreted as a direct comparison of human and model reliability under a common specification. Re-extracting the round-four speeches using the complete revised prompt would be required for such a comparison. Uncertainty attention also remains comparatively difficult to classify consistently. Despite the targeted clarification introduced before round four, its estimated κ remains low and its confidence interval includes zero. This does not establish that the construct is intrinsically unreliable, but it indicates that the current operational definition may require further conceptual refinement.

Finally, comparisons across validation rounds are not controlled experiments. Successive rounds differ in several dimensions, including sample composition, sampling design, information available to the coders, and the coding rules themselves. Improvements in agreement across rounds therefore cannot be attributed exclusively to any single modification. The validation sequence is better interpreted as an iterative diagnostic process: each round identifies limitations in the preceding design and motivates changes that are subsequently evaluated, rather than providing a sequence of directly comparable estimates of classification performance.

5 Misclassification Correction

The attention measures observed over the full corpus are model classifications rather than human reference codes. If the model produces false positives or false negatives, the observed share of speeches classified as discussing a topic may therefore differ from the share that would be obtained under the reference classification. Human coding is available only for the validation samples and cannot replace the model classifications over the full corpus. Instead, the validation exercise can be used to estimate the model’s misclassification rates and, under assumptions about their applicability to the remaining speeches, to infer the underlying population share consistent with the observed model classifications. Misclassification of observed categorical outcomes is a standard measurement-error problem and can distort subsequent statistical analysis when the observed classification differs systematically from the underlying reference state (Hausman et al., 1998).

¹Appendix Section A.3

The correction below illustrates this procedure using the sensitivity and specificity estimates obtained in the initial validation round. Subsequent validation showed that these parameters are sensitive to the validation design and to the choice of human reference classification. The numerical results reported in this section should therefore be interpreted as an illustration of the correction procedure rather than as preferred estimates of corrected attention levels.

5.1 Correction Framework

Let $D^* \in \{0, 1\}$ denote the true code for a speech and D the code the instrument assigns. Write $\alpha = \Pr(D = 1 \mid D^* = 1)$ for sensitivity and $\beta = \Pr(D = 0 \mid D^* = 0)$ for specificity, and let $p = \Pr(D^* = 1)$ be the population share of interest. By the law of total probability,

$$\Pr(D = 1) = \alpha p + (1 - \beta)(1 - p), \quad (3)$$

which rearranges to

$$p = \frac{\Pr(D = 1) - (1 - \beta)}{\alpha + \beta - 1}. \quad (4)$$

Intuitively, the observed model-positive share contains two groups: reference-positive speeches correctly identified by the model and reference-negative speeches incorrectly classified as positive. Equation (4) reverses this relationship. Starting from the positive classification rate observed over the full corpus, it uses the estimated sensitivity and specificity to recover the underlying positive share that would be consistent with the observed model classifications.

Replacing the population quantities by sample counterparts gives the estimator implemented in `src/measurement_error.py`. Appendix Section A.4 derives the correction in full. The inversion requires $\alpha + \beta > 1$, corresponding to the requirement that the classifier contain information about the reference classification in the intended direction.

The transformation in Equation (4) is not simply a constant adjustment to the observed share. For fixed sensitivity and specificity, it removes the component associated with false positives and rescales the resulting quantity to account for the classifier’s error rates. Because these rates are themselves estimated, the correction can produce values outside the admissible interval $[0, 1]$ in finite samples. The implementation clips such values to the unit interval for interpretability; this is a practical restriction rather than part of the underlying inversion.

5.2 Application of the Misclassification Correction

Table 9 applies Equation (4) using the error-rate estimates from the initial validation round. These values are retained to illustrate how estimated classification error propagates into aggregate attention shares and should not be interpreted as final corrected estimates. In the initial validation sample, sensitivity was estimated at one for every attention variable. Under these estimates, the adjustment therefore operates through estimated false positives. For financial stability and uncertainty, specificity was also estimated at one, so their point estimates are unchanged.

Table 9: Misclassification-corrected attention shares

Variable	Raw share	Corrected	Conservative 95% bounds
Inflation attention	0.588	0.515	[0.336, 0.843]
Growth attention	0.798	0.765	[0.647, 1.000]
Financial stability attention	0.702	0.702	[0.583, 0.853]
Uncertainty attention	0.549	0.549	[0.369, 0.666]

The width of the reported bounds reflects uncertainty in the estimated misclassification rates. The bounds are propagated from the Clopper–Pearson intervals for sensitivity and specificity through Equation (4). With only thirty speeches in the initial validation sample, the error rates are estimated imprecisely, and this uncertainty translates into wide ranges for the implied corrected shares. The table

therefore illustrates an important feature of the procedure: observing the model-positive share precisely over a large corpus does not eliminate uncertainty about the underlying share when the misclassification parameters are estimated from a much smaller validation sample.

Figure 2 shows the same correction applied to the annual attention series.

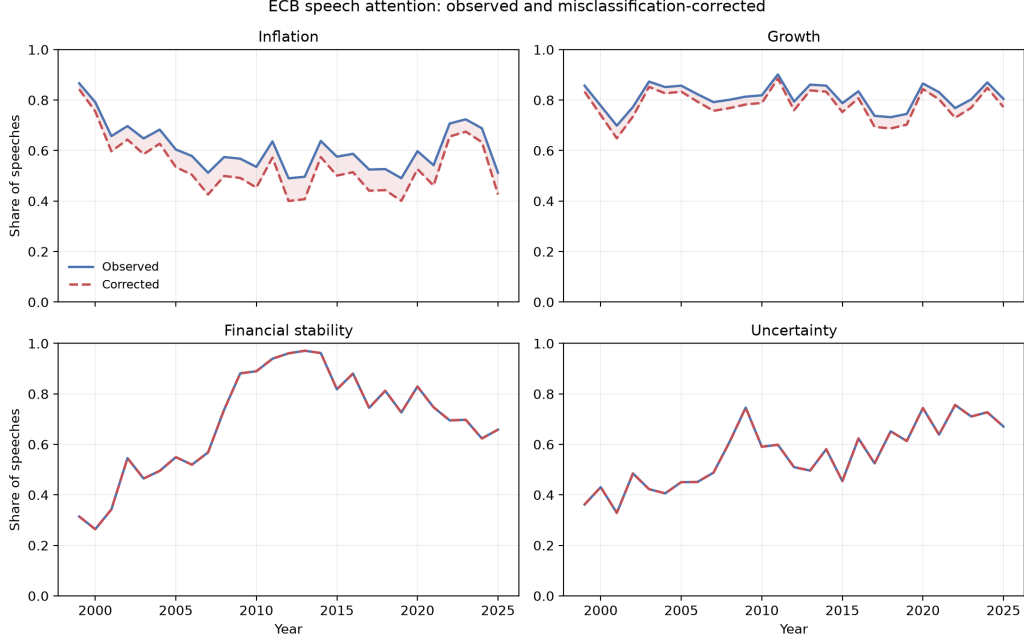


Figure 2: Observed and misclassification-corrected annual attention shares.

Notes: The figure compares observed annual attention shares with shares obtained from the binary misclassification correction in Equation (4). The correction uses the sensitivity and specificity estimates from the initial validation round and is therefore illustrative rather than a preferred estimate of the underlying attention series. The adjustment is not a constant level shift: with sensitivity equal to one and specificity below one, the difference between the observed and corrected shares is larger when observed attention is lower.

5.3 Interpretation and Limitations

The parameters used in the illustrative correction are those estimated in round one and retained in `outputs/validation/measurement_error_params.json`. The subsequent validation rounds provide reasons not to interpret these parameters as definitive estimates of the model’s error process. Round one relied on a small, single-coder validation sample and did not permit truncation-induced classification errors to be detected. Later rounds also showed that estimated model error depends on the human reference classification and that some disagreement reflected ambiguities and differences in the coding specifications.

Round four provides a stronger assessment of inter-coder reproducibility under the revised coding scheme, but it does not provide a definitive replacement set of model sensitivity and specificity parameters. The revised human codebook is more closely aligned with the production prompt used for the existing extraction, but it also contains six additional decision rules introduced after that extraction was completed. A clean re-estimation of the model’s misclassification rates would therefore require applying the complete revised specification to the model and comparing the resulting classifications with human codes produced under the same rules. Until such an exercise is conducted, Table 9 and Figure 2 should be read as demonstrations of the correction procedure rather than estimates of the underlying attention levels.

The correction itself addresses a specific measurement problem: bias in the prevalence of a binary classification arising from false positives and false negatives. It does not eliminate uncertainty in the

estimated sensitivity and specificity parameters, nor does the current analysis propagate that parameter uncertainty through quantities constructed downstream. The correction also does not apply directly to the conditional outlook measures, which take three values and are computed over speeches selected by the corresponding attention classification. Correcting those measures would require a richer characterization of the joint misclassification process.

Finally, applying a common sensitivity and specificity across the corpus assumes that the classifier’s error process is sufficiently stable across time and relevant speech characteristics. If classification performance varies systematically with characteristics such as speech length, language, period, or communication style, a single pair of error rates may not adequately characterize measurement error in every period. A larger validation sample would be required to investigate such heterogeneity systematically.

6 From Speech-Level Classifications to Communication Measures

The extraction procedure produces classifications at the level of the individual speech. To use these classifications in subsequent economic analysis, they are aggregated into time-series measures of ECB communication. The aggregation reflects a distinction between three dimensions: the prevalence with which a topic is discussed, the number of speeches in which it receives substantive attention, and the direction of the outlook conditional on that attention.

6.1 Construction of the Measures

For topic x in period t containing N_t speeches indexed by i , the aggregation computes

$$\text{Attention}_t^x = \frac{1}{N_t} \sum_i A_i^x, \quad (5)$$

$$\text{Volume}_t^x = \sum_i A_i^x, \quad (6)$$

$$\text{Outlook}_t^x = \frac{\sum_i A_i^x O_i^x}{\sum_i A_i^x}, \quad (7)$$

where A_i^x is the attention indicator and O_i^x the corresponding outlook classification.

Attention measures the share of speeches in a period that substantively discuss topic x , while volume records the number of such speeches. The distinction separates the prevalence of a topic across ECB speeches from changes in the overall amount of communication. Two periods can therefore have the same attention share while differing substantially in the absolute number of speeches discussing the topic.

The outlook measure captures the average directional classification among speeches that substantively discuss the topic. Since A_i^x is binary, Equation (7) is equivalent to taking the mean of O_i^x over speeches for which $A_i^x = 1$. Positive and negative values indicate that the conditional distribution of outlook classifications is tilted towards the corresponding direction, while values close to zero indicate a predominantly neutral or approximately balanced set of directional classifications.

Keeping attention and outlook separate allows changes in the prevalence of a topic to be distinguished from changes in its directional assessment. A period in which inflation is discussed frequently but without a dominant directional outlook is therefore distinct from one in which inflation receives relatively little attention but the speeches that address it consistently emphasize the same direction. This distinction is also relevant for comparison with simpler text-based measures. A single score based on the occurrence of directional terms may vary because the relevant topic receives greater textual attention, because the language conditional on that discussion becomes more directional, or because both margins change. Constructing attention and outlook separately makes these two components observable as distinct quantities. This distinction is also economically relevant: attention captures which topics receive emphasis in central-bank communication, while the conditional outlook captures their stated expected direction and can therefore be related more directly to expectations, forward guidance, and the anticipated path of economic conditions. More generally, treating central-bank communication as

multidimensional rather than reducing it to a single textual score is consistent with existing text-based work that separates distinct informational components of monetary-policy communication (Hansen and McMahon, 2016).

The communication measures are constructed at monthly, quarterly, and annual frequencies. Quarterly aggregation is used for the main analysis as a compromise between temporal resolution and the number of speeches available within each period. Of the 324 months in the sample, 35 contain fewer than three speeches and two contain no speeches, making monthly attention shares and conditional outlook measures potentially unstable or, where the relevant denominator is zero, undefined. By contrast, every quarter contains at least three speeches. Quarterly aggregation therefore provides a more stable basis for constructing the communication measures while retaining greater temporal resolution than annual aggregation.

6.2 Dispersion and Polarization

The aggregation additionally records variation in outlook classifications across speeches. Dispersion is measured as the standard deviation of outlook codes among speeches for which the corresponding attention indicator equals one. This provides information that the mean outlook alone cannot capture: an average close to zero may arise because most speeches are neutral or because positive and negative directional assessments offset one another, as stated above.

A complementary measure of directional disagreement is defined as

$$\text{Polar}_t^x = \frac{2 \min(u_t, d_t)}{u_t + d_t}, \quad (8)$$

where u_t and d_t denote the number of speeches classified as +1 and −1, respectively, among those that substantively discuss topic x . The measure equals zero when all directional classifications point in the same direction and one when positive and negative classifications are equally represented. Neutral classifications are excluded, so polarization captures the balance between opposing directional assessments rather than the overall frequency of neutral communication.

Dispersion and polarization exploit the cross-sectional variation available in speech-level classifications and therefore contain information that is not summarized by the conditional mean outlook. They are constructed as part of the measurement framework but are not used in the subsequent empirical analysis.

6.3 Summary of the Communication Measures

Table 10 summarizes the quarterly communication measures over the 108 quarters from 1999Q1 to 2025Q4. Growth receives the highest average attention, with a quarterly mean of 0.813 and a minimum of 0.500. Financial-stability attention exhibits considerably greater variation, ranging from 0.125 to 1.000, while inflation and uncertainty attention occupy intermediate ranges. The conditional outlook measures vary on both sides of zero: inflation outlook ranges from −0.833 to 0.917, with a mean of −0.076, while growth outlook ranges from −0.769 to 0.929, with a mean of 0.147. Polarisation also varies substantially over time, with some quarters characterized by broad agreement in the direction of the outlook and others by a more even division between positive and negative assessments.

For completeness, the table also reports the misclassification-corrected inflation-attention series constructed in the preceding section. As discussed in the previous section, this series uses the round-one error-rate estimates and is retained as an illustration of the correction procedure rather than as a preferred estimate of the underlying attention level.

6.4 Distinct Information in Attention and Outlook

The construction above treats attention and conditional outlook as distinct dimensions of communication. Attention measures how frequently growth receives substantive discussion, whereas outlook measures the direction of the assessment conditional on growth being discussed. The distinction is economically meaningful only if the two measures contain different information. If instead they were simply two

Table 10: Quarterly communication measures, 1999Q1–2025Q4 (108 quarters)

Measure	Mean	S.D.	Min	Max
Speeches per quarter	25.16	8.70	8.00	51.00
Inflation attention	0.611	0.135	0.312	0.958
corrected	0.542	0.159	0.191	0.951
Growth attention	0.813	0.094	0.500	1.000
Financial stability attention	0.694	0.218	0.125	1.000
Uncertainty attention	0.560	0.153	0.143	0.867
Inflation outlook	−0.076	0.373	−0.833	0.917
Growth outlook	0.147	0.352	−0.769	0.929
Inflation polarization	0.383	0.331	0.000	1.000
Growth polarization	0.525	0.295	0.000	1.000

noisy measures of the same underlying communication stance, separating them would add little to the analysis.

This can be examined by comparing the two measures against the same external economic variable. Both growth measures were pre-specified for comparison with the change in euro-area unemployment over the following two quarters, defined such that a positive change represents rising unemployment. The expected relationships differ because the two measures capture different dimensions of communication. Greater growth attention was expected to be positively associated with subsequent unemployment changes, on the hypothesis that real activity receives greater attention when concerns about economic weakness become more salient. A more positive growth outlook was instead expected to be negatively associated with subsequent unemployment changes, since an improving expected path of economic activity should be associated with lower unemployment.

Using the same target, sample, and estimator for both measures provides a direct comparison of their empirical content. Specifically, the exercise compares the correlation between growth attention and subsequent unemployment changes with the corresponding correlation for growth outlook, and tests whether the two correlations differ. Unlike a comparison across alternative classification methods, the underlying instrument and corpus are held fixed; only the communication measure being evaluated changes.

Table 11: Growth attention against growth outlook, same target and sample

Signal	r	HAC s.e.	Difference	p
Growth attention	+0.060	0.102	0.409	0.032
Growth outlook	−0.349	0.145		

The target is the change in euro-area unemployment over the following two quarters, measured over 102 quarters. The first two columns report the correlation of each communication measure with this common target and its Newey–West HAC standard error. The final two columns test whether the two correlations are equal. Because both correlations use the same unemployment series and observations, their sampling errors are not independent; the test therefore estimates the difference directly using standardised cross-products. The estimated difference is 0.409, with a 95% confidence interval of $[+0.040, +0.779]$ and $p = 0.032$. HAC inference uses four lags to account for serial dependence, including that induced by the overlapping two-quarter outcome. The comparison was pre-specified, so no multiplicity adjustment is applied.

The results indicate substantially different relationships with the same external target. Growth attention exhibits little association with subsequent unemployment changes: its estimated correlation is +0.060 with a HAC standard error of 0.102, and is not statistically distinguishable from zero. Growth

outlook, by contrast, has an estimated correlation of -0.349 with a HAC standard error of 0.145 . The negative sign is consistent with the pre-specified direction: more positive growth-outlook classifications are associated with lower changes in unemployment over the following two quarters.

The difference between the two estimated correlations is 0.409 . Because both correlations contain the same unemployment variable, they cannot be treated as independent estimates. The difference is therefore tested directly as the mean difference between the corresponding standardized cross-products, with Newey–West inference accounting for serial dependence. The resulting p -value is 0.032 , with a 95% interval of $[+0.040, +0.779]$. The data therefore reject equality of the two correlations at the 5% level.

The two communication measures are also only weakly related to one another, with a correlation of -0.081 across the same quarters. This provides additional evidence that the prevalence of growth discussion and the direction of the conditional growth outlook are not interchangeable measures of a single communication dimension.

Taken together, these results support the decision to maintain attention and outlook as separate quantities in Equations (5) and (7). Intuitively, attention measures how frequently growth receives substantive discussion across ECB speeches, while outlook measures the average expected direction of economic activity among the speeches in which growth is discussed. In this sample, the former has little linear association with subsequent unemployment changes, whereas the latter exhibits a moderate negative association with the same target. Collapsing the two dimensions into a single score could therefore obscure differences between changes in how often the ECB discusses growth and changes in what direction it communicates when doing so. Whether a simpler dictionary-based classification method preserves these distinctions is examined separately in Section 8.

7 External Validation Against Euro-Area Macroeconomic Data

7.1 Validation Strategy

The internal validation assesses the extent to which the model reproduces human classifications under the coding scheme. Agreement with human coding, however, is not sufficient to establish that the resulting measures have a meaningful economic interpretation. A classifier could apply a coding scheme consistently while generating measures that bear little relationship to observable economic conditions.

The external validation therefore asks whether the constructed communication measures are associated with macroeconomic variables in directions consistent with their intended interpretation. The analysis in this section is one of concordance and construct validity rather than causal identification. Most comparisons relate communication measures to macroeconomic conditions observed in the same quarter and are therefore interpreted as contemporaneous measures of concordance. The growth measures are additionally compared with the change in unemployment over the following two quarters to assess whether their forward-looking content is associated with subsequent labour-market developments. The latter comparison remains an external-validation exercise rather than a forecasting test: it is conducted in-sample and does not assess whether the communication measures provide incremental or out-of-sample predictive power. Similarly, none of the associations identifies a causal effect of ECB communication on economic outcomes.

7.2 External Validation Data

Four external series are obtained from the ECB Data Portal. The HICP annual inflation rate, `ICP/M.U2.N.000000.4.ANR`, is observed monthly and averaged within each quarter. The primary inflation target is the absolute deviation from the two per cent reference, $|\text{HICP} - 2|$, on the hypothesis that inflation receives greater attention when realized inflation is further from the reference level in either direction. The signed deviation is retained as a robustness variant.

The euro-area unemployment rate, `LFSI/M.I9.S.UNEVRT.TOTALO.15_74.T`, is also observed monthly and averaged within the quarter. The series begins in 2000Q1 and is therefore unavailable for the first four quarters of the speech sample. The main target for the growth measures is the change in unemployment over the following two quarters, while the contemporaneous level is retained as an alternative operationalization.

Financial-stability attention is compared with the Composite Indicator of Systemic Stress (CISS) (Hollo et al., 2012), `CISS/D.U2.ZOZ.4F.EC.SS_C1.IDX`, aggregated to quarterly frequency. The original CISS is retained as the primary measure because it was specified before estimation, while the daily New CISS (`SS_CIN`) is used as a robustness variant. Financial-stability attention is expected to increase with the CISS, which is constructed to summarize systemic stress across different segments of the financial system (Hollo et al., 2012).

Uncertainty attention is compared with cross-forecaster dispersion in Survey of Professional Forecasters point forecasts for real GDP growth, `SPF/Q.U2.RGDP.POINT.P9M.Q.VAR`. Forecast disagreement provides an observable proxy for uncertainty, although disagreement and aggregate forecast uncertainty need not coincide, particularly when uncertainty associated with common shocks is important (Lahiri and Sheng, 2010). The published quantity is a variance in squared percentage points, so its square root is taken to express dispersion in the same units as the underlying forecasts. The published period identifies the end of the forecast horizon rather than the survey round in which forecasts were submitted. The series is therefore shifted back by the corresponding horizon to align dispersion with the quarter in which forecasters expressed their views: two quarters for the nine-month horizon and three for the twelve-month horizon. Real GDP dispersion is the primary uncertainty proxy, while HICP forecast dispersion is retained as a robustness variant.

7.3 Statistical Inference

Inference accounts for serial dependence in both the communication measures and the macroeconomic variables. This is particularly important for the unemployment target, which is constructed as an overlapping two-quarter change and therefore induces additional dependence between adjacent observations. Correlations are estimated through regressions of standardized variables with Newey–West heteroskedasticity- and autocorrelation-consistent standard errors (?). Four lags are used at the available sample sizes, with the bandwidth constrained to address the dependence induced by overlapping outcomes. Moving-block bootstrap intervals provide a non-parametric robustness check (Künsch, 1989).

Every estimate in this section can be represented within the same specification. Write $\tilde{z}$ for standardization to zero mean and unit variance over the estimation sample, m for a communication measure, y for its target, and x for an optional vector of controls:

$$\tilde{y}_t = \alpha + \beta \tilde{m}_t + \gamma' \tilde{x}_t + \varepsilon_t, \quad L = \max\left\{\left\lfloor 4(T/100)^{2/9} \right\rfloor, h - 1\right\}. \quad (9)$$

The coefficient β is estimated by OLS, with Newey–West standard errors using Bartlett bandwidth L . When no controls are included, $\hat{\beta}$ is the sample correlation between m and y , since both variables are standardized. The five principal comparisons are estimated in this form, so the correlations reported in Table 12 correspond to the coefficients from Equation (9), with HAC inference applied to account for serial dependence.

The same specification is used for the additional growth-outlook exercise by including the contemporaneous unemployment level in $\tilde{x}_t$. In this case, y is the two-quarter change in unemployment and m is growth outlook. The coefficient $\hat{\beta}$ therefore measures the association between growth outlook and the subsequent unemployment change conditional on the contemporaneous unemployment level, rather than the unconditional correlation between the two variables.

The horizon of the target also enters the bandwidth rule. For contemporaneous targets, $h = 1$. The term $\lfloor 4(T/100)^{2/9} \rfloor$ is a conventional sample-size-based rule of thumb for selecting the Newey–West bandwidth, where T is the number of observations in the estimation sample. The bandwidth determines how many lagged autocovariances are included in the HAC variance estimator and increases gradually with the length of the time series. For the unemployment target, the two-quarter horizon implies $h = 2$, and the bandwidth is constrained to be at least $h - 1$ to accommodate the dependence generated by overlapping observations. At the sample sizes used here, approximately $T = 100$ quarterly observations, the automatic component gives $4(T/100)^{2/9} \simeq 4$ and therefore exceeds the overlap-based lower bound, resulting in $L = 4$ throughout. Thus, the same estimation procedure is applied to all five principal comparisons. Only the communication measure and external target (m, y), together with the sample of quarters for which both are available, differ across specifications; the standardization, estimator, and Newey–West bandwidth rule are otherwise held constant.

Five principal comparisons were specified before estimation, including the communication measure, external target, and expected sign. Holm’s step-down correction (Holm, 1979) is applied across these five tests to control the family-wise error rate, and adjusted p -values are reported alongside the unadjusted values. Four alternative constructions are reported separately as robustness checks rather than as additional primary hypotheses.

7.4 Pre-Specified Validation Relationships

Table 12: Salience pairs, signs fixed before estimation

Measure	Target	Sign	n	r	s.e.	p	p_{Holm}
Uncertainty attention	SPF GDP dispersion	+	106	+0.361	0.163	0.028	0.114
Growth outlook	Unemployment, 2q change	−	102	−0.349	0.145	0.018	0.091
Financial stability attention	CISS	+	106	+0.246	0.144	0.092	0.277
Inflation attention	$ \text{HICP} - 2 $	+	108	+0.167	0.167	0.319	0.637
Growth attention	Unemployment, 2q change	+	102	+0.060	0.102	0.555	0.637

Newey–West, four lags throughout. *Sign* is the direction fixed in advance. Holm applied across these five.

The expected signs follow from the intended economic interpretation of each communication measure. Uncertainty attention is expected to be positively associated with SPF forecast dispersion: greater disagreement across professional forecasters indicates a wider range of views about the economic outlook and should therefore coincide with greater emphasis on uncertainty in ECB communication. Financial-stability attention is expected to increase with the CISS because higher systemic stress reflects greater strain and fragility in the financial system, making financial-stability risks more likely to receive substantive attention in ECB speeches. Inflation attention is expected to increase with the absolute deviation of inflation from the two per cent reference, since larger deviations in either direction make price developments more salient for a central bank whose mandate is centered on price stability. Growth outlook is expected to be negatively associated with the subsequent two-quarter change in unemployment: a more positive expected trajectory for economic activity should be associated with improving labour-market conditions and therefore a decline, or smaller increase, in unemployment. Growth attention is instead expected to be positively associated with subsequent unemployment changes, under the hypothesis that real activity becomes more salient in ECB communication when concerns about economic weakness and deteriorating labour-market conditions increase. The expected sign for growth attention is less direct. Greater attention to economic activity may arise when growth is weakening, but it may also accompany a strong recovery. The positive sign therefore reflects the specific pre-specified hypothesis that growth becomes more salient in ECB communication during periods of economic weakness, rather than a general prediction of economic theory.

All five estimated correlations have the pre-specified sign. The largest correlations are observed for uncertainty attention and SPF GDP dispersion, at +0.361, and for growth outlook and the subsequent two-quarter change in unemployment, at −0.349. Two associations have unadjusted p -values below 0.05, but none survives the Holm adjustment; the smallest adjusted value is 0.091. The results therefore provide directional consistency with the pre-specified hypotheses, while no individual association should be regarded as established.

7.5 Inflation Attention and Distance from the Reference Level

Inflation attention provides an informative weak result because the measure is comparatively well reproduced by the human coders and exhibits substantial time-series variation. Round-four human–human agreement is $\kappa = 0.800$, while quarterly inflation attention has a mean of 0.611, a standard deviation of 0.135, and ranges from 0.31 to 0.96. Nevertheless, its correlation with the absolute distance of inflation from 2% is only +0.167, with $p = 0.319$. The data therefore provide little evidence of a strong linear relationship between the prevalence of inflation discussion and the distance of realized inflation from the reference level. The relatively high inter-coder reliability and substantial variation in the measure make classification noise or limited variation less convincing explanations for the weak association, although neither can be ruled out completely.

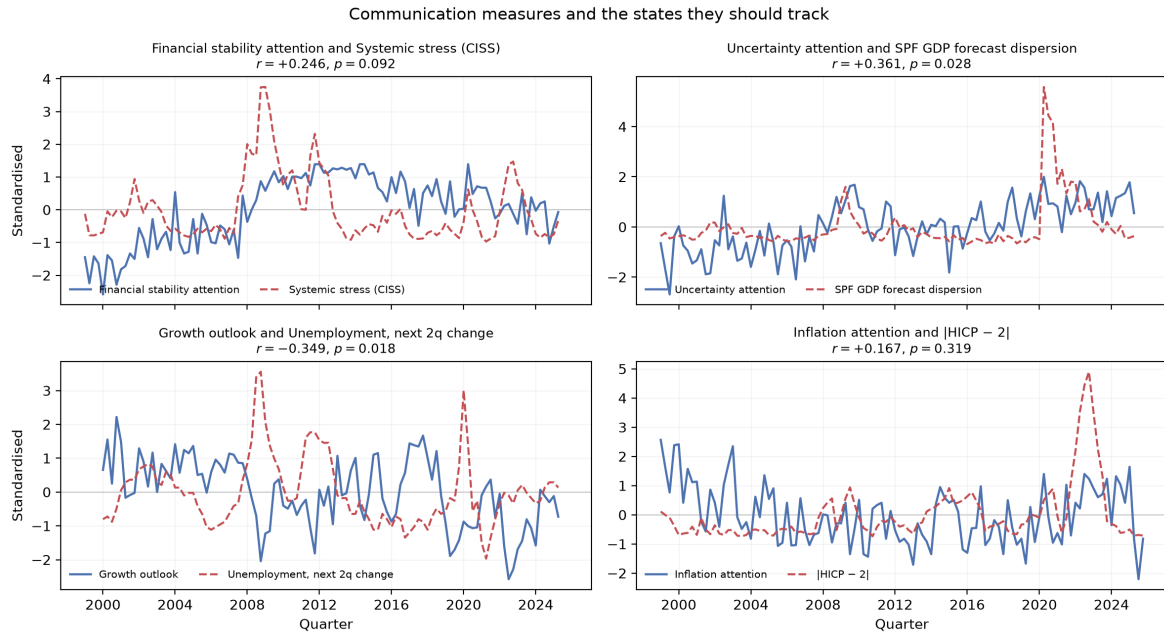


Figure 3: Communication measures and their corresponding external validation variables.

Notes: Each panel plots a quarterly communication measure and its corresponding external variable after standardization to mean zero and unit variance over the estimation sample. Reported r values are Pearson correlations. Reported p -values are based on Newey–West heteroskedasticity- and autocorrelation-consistent standard errors with four lags. The unemployment target is the subsequent two-quarter change and therefore overlaps across adjacent observations. Sample sizes vary across panels according to data availability.

7.6 Growth Outlook and Unemployment

The relationship between growth outlook and unemployment is examined more closely because it is the strongest individual association in Table 12 and admits a potential mechanical explanation. Growth outlook correlates +0.241 with the contemporaneous level of unemployment. A potential concern is that this relationship simply reflects the initial level of unemployment. If more positive growth outlooks tend to occur when unemployment is high, and high unemployment subsequently tends to decline through mean reversion, growth outlook would appear to be associated with falling unemployment even if it contains no information beyond the contemporaneous unemployment level.

The data provide little evidence for this mechanism. The contemporaneous unemployment level correlates -0.041 with its own two-quarter change, and controlling for the level leaves the coefficient on growth outlook essentially unchanged: -0.360 with $p = 0.015$, compared with -0.349 and $p = 0.018$ without the control. The negative association between growth outlook and subsequent unemployment changes therefore remains after conditioning on the contemporaneous unemployment level. Given the Holm-adjusted results, this relationship remains suggestive rather than established; the exercise rules out one simple mechanical explanation without establishing a causal or forecasting relationship.

Table 13: Robustness variants, outside the Holm family

Pair	Change	n	r	p
Uncertainty attention	Inflation rather than GDP dispersion	105	+0.349	0.037
Financial stability attention	New CISS rather than original	108	+0.227	0.137
Inflation attention	Signed rather than absolute gap	108	+0.200	0.208
Growth attention	Unemployment level rather than change	104	+0.147	0.087

Each alternative construction preserves the sign and approximate magnitude of its pre-specified counterpart, and none changes the qualitative interpretation of the corresponding primary comparison. This provides evidence that the main conclusions are not especially sensitive to reasonable alternative definitions of the external validation variables.

7.7 Robustness to Speech Length

Speech length is a potential confound for the attention measures because longer speeches provide more opportunities for a topic to receive substantive discussion, while average speech length changes substantially over the sample. To assess whether this feature drives the external-validation results, each communication measure is residualized with respect to the two speech-length controls used in the analysis and the external correlations are recomputed. The two controls are themselves strongly correlated, at +0.835, and should therefore be interpreted primarily as capturing a common dimension of speech length rather than as two independent sources of variation.

Table 14: External validation after controlling for speech length

Measure	Target	n	Raw r	Length-adjusted r	Change
Financial stability attention	CISS	106	+0.246	+0.332	+0.086
Uncertainty attention	SPF GDP dispersion	106	+0.361	+0.352	-0.009
Growth attention	Unemployment, 2q change	102	+0.060	+0.050	-0.010
Growth outlook	Unemployment, 2q change	102	-0.349	-0.368	-0.019
Inflation attention	HICP - 2	108	+0.167	+0.182	+0.015

Length-adjusted values are semi-partial correlations obtained after residualizing the communication measure with respect to the two speech-length controls. Newey–West inference uses the same bandwidth rule as the primary specifications. Full partial correlations, obtained by additionally residualizing the external target, are nearly identical.

The adjustment leaves the main results essentially unchanged. All five correlations retain their original signs, no individual test crosses the 5% significance threshold in either direction, and four of

the five correlations change by less than 0.02 in absolute value. Applying the Holm correction across the five adjusted tests reduces the smallest adjusted p -value from 0.091 in the baseline results to 0.055, which remains above the 5% family-wise threshold. The conclusion of the external-validation exercise is therefore not sensitive to the association between speech length and the constructed communication measures.

The largest change occurs for financial-stability attention, whose correlation with the CISS increases from 0.246 to 0.332 after the length adjustment. Financial-stability attention is negatively correlated with mean log speech length, at -0.467 : speeches become shorter over the sample while financial-stability attention increases particularly after the financial crisis. The common length trend therefore attenuates rather than generates the positive association with systemic stress, and controlling for it strengthens the external-validation relationship. Finally, residualizing both the communication measure and the external target produces correlations that are nearly identical to the semi-partial results, so the choice between the two adjustment procedures has no substantive effect on the conclusions.

7.8 Interpretation and Limitations

Overall, the results are broadly consistent with the intended economic interpretation of the measures, but they provide limited statistical evidence of external validity. All five primary correlations have the signs specified before estimation, and the alternative constructions preserve those directions. At the same time, no individual association survives adjustment across the five primary tests. The evidence therefore supports the narrower conclusion that the constructed measures exhibit relationships with external economic variables that are broadly consistent with their intended interpretation, rather than establishing any individual relationship as a robust empirical regularity.

The exercise does not support a causal interpretation. For the contemporaneous comparisons, the communication measures and economic conditions are observed within the same quarter. The resulting correlations therefore establish co-movement but do not identify its direction: ECB communication may respond to prevailing economic conditions, both may respond to common developments, or other mechanisms may generate the observed association. The growth measures are compared with a subsequent two-quarter unemployment change, but these associations do not constitute an out-of-sample forecasting exercise and do not establish independent predictive content.

Two additional limitations concern measurement. First, the communication measures enter the analysis as observed regressors even though they are outputs of a classification procedure. Uncertainty from the extraction and validation stages is not propagated through the reported standard errors, which are therefore conditional on the constructed measures being taken as given. Second, speech length may affect the attention measures because longer speeches provide more opportunities for a topic to receive substantive discussion, while average speech length declines by approximately 26% over the sample period. This concern is less direct for the conditional outlook measures, although attention classification still determines which speeches enter those measures.

8 Benchmarking Against a Keyword Dictionary

8.1 Benchmarking Strategy

A simpler alternative to LLM-based classification is to construct the communication measures from predefined keywords and directional terms. Dictionary-based methods provide a transparent approach to converting text into quantitative measures, but their performance depends on whether the lexical categories correspond to the meaning of terms in the relevant domain (Loughran and McDonald, 2011). The benchmark therefore asks whether the additional contextual information used by the language model produces classifications that differ meaningfully from those generated by a keyword-based classification method. The comparison concerns both agreement with human coding and the economic content of the resulting measures: two methods may differ not only in how consistently they reproduce human classifications, but also in which features of the underlying text their measures capture.

8.2 Construction of Dictionary Benchmark

The dictionary is designed as an implementation of the same measurement framework rather than as a simple baseline. Three features of the comparison are held approximately constant. First, the dictionary is applied to the same truncated excerpts supplied to the language model, so differences between the two methods cannot be attributed to differences in the text observed. Second, its vocabulary is constructed from the concepts identified as qualifying evidence in the coding scheme, translating the same substantive definitions into lexical rules. This domain-specific construction is motivated by evidence that generic dictionaries can systematically misclassify terms when their meaning differs across contexts (Loughran and McDonald, 2011). Third, the dictionary thresholds are calibrated to reproduce the model’s classification rates. For example, if the model classifies 60% of speeches as receiving substantive inflation attention, the dictionary threshold is chosen so that approximately 60% of speeches are also classified as positive. This separates differences in which speeches are classified from differences arising simply because one method assigns positive classifications more frequently than the other. The calibrated dictionary classification rates differ from the corresponding model rates by no more than 0.008 for any variable except financial stability. For that variable, several speeches receive the same dictionary score around the classification threshold, preventing an exact match and leaving a difference of 0.044.

The benchmark also incorporates two contextual refinements. Directional terms contribute to the outlook score only when they occur within sentences that mention the corresponding topic, and a three-word negation window reverses the sign assigned to negated directional terms. The dictionary therefore incorporates limited information about local context rather than relying on unrestricted document-level word counts. Matching its classification rates to those of the model additionally gives the benchmark information that a standalone dictionary would not ordinarily possess, making the comparison deliberately favorable to the dictionary based classification.

8.3 Agreement with Human Coding

Both methods are first evaluated against the same thirty manually coded speeches from the initial validation round, using an identical scoring procedure. Across the 180 speech–variable classifications, model–human agreement is $170/180 = 0.944$, compared with $131/180 = 0.728$ for the dictionary. These figures should be interpreted as relative agreement with a common human reference rather than definitive estimates of classification accuracy, given the limitations of the round-one human coding documented earlier.

The aggregate difference also varies substantially across variables. Inflation attention produces an exact tie: agreement is 0.900 with Cohen’s $\kappa = 0.791$ for both methods, and both generate three false positives, two of them on the same speeches. This suggests that lexical information captures much of the distinction required to identify substantive inflation discussion, leaving relatively little advantage for contextual classification in this particular task. Growth attention produces a different result, with dictionary specificity at 0.357, compared with 0.857 for the model, indicating substantially more false-positive classifications under the lexical procedure. Real-activity vocabulary is common in central-bank communication even when growth is not the substantive focus of a speech, making the distinction between the occurrence of relevant terms and substantive discussion more important.

Uncertainty presents a different classification problem. The dictionary misses six of the nineteen human-positive observations. The coding scheme explicitly requires more than the occurrence of generic terms such as “risk”: uncertainty must be substantively emphasized in the sense defined by the classification rules. A lexical procedure has limited ability to distinguish these uses from routine references to risks, whereas the LLM classification can use the surrounding context. The benchmark therefore suggests that the relative advantage of contextual classification is heterogeneous across variables and is greatest where the coding decision depends on how terms are used rather than simply whether they occur.

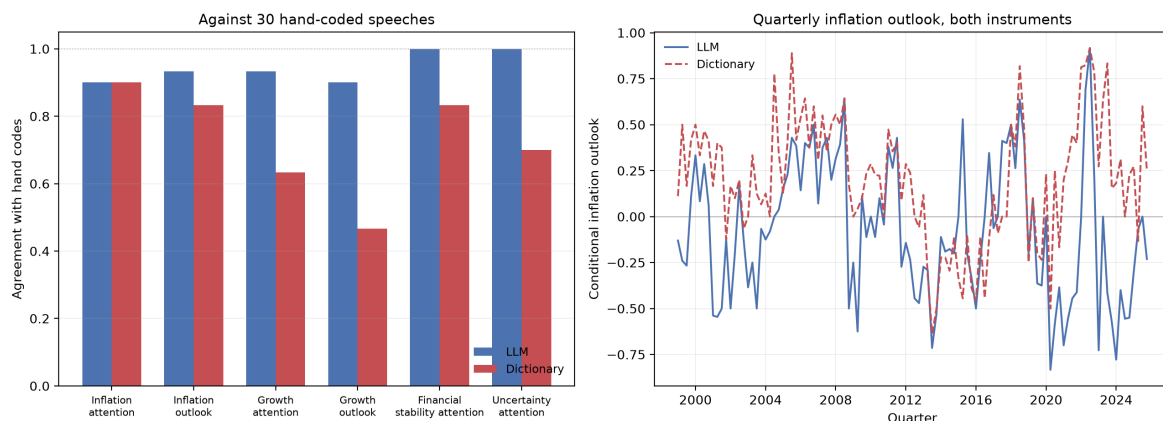


Figure 4: Comparison of the language-model and dictionary-based classifications.

Notes: Left panel: agreement rates of the LLM and dictionary classifications with the same 30 hand-coded speeches, reported separately by variable. Both methods are evaluated on the same speech excerpts and against the same human reference classifications. Right panel: quarterly conditional inflation-outlook measures constructed from the LLM and dictionary classifications. Conditional outlook is averaged over speeches for which the corresponding inflation-attention indicator equals one. The correlation between the two quarterly outlook series is 0.482.

8.4 External Comparison of the Outlook Measures

A comparison with realized inflation initially produces the opposite ranking. The dictionary inflation-outlook measure correlates 0.618 with the contemporaneous HICP level, compared with 0.229 for the model measure, and the dictionary correlation is the only one of the two that is statistically significant at conventional levels. If performance were evaluated solely by the strength of the relationship with realized inflation, the simpler method would therefore appear to perform better.

This comparison, however, must be interpreted relative to the construct the outlook variable is intended to measure. The coding scheme instructs the classifier to distinguish the expected direction of inflation from its current level. A strong relationship with contemporaneous HICP may therefore reflect sensitivity to the level of inflation rather than successful measurement of directional outlook. More generally, dictionary-based text measures can exhibit systematic relationships with external variables when lexical categories capture features of the text other than the construct they were intended to measure (Loughran and McDonald, 2011). To examine this possibility, the two measures are compared separately with the contemporaneous HICP level, the change in HICP over the preceding quarter, and the change over the following two quarters.

Table 15: What the two outlook measures track

Target	Language model	Dictionary
HICP level	0.229 (0.150)	0.618 (0.187)
HICP change, past quarter	0.338 (0.144)	0.268 (0.140)
HICP change, next two quarters	−0.011 (0.121)	−0.041 (0.154)

Newey–West standard errors in parentheses, 108 quarters.

The two measures exhibit different patterns across these targets. The dictionary’s strongest relationship is with the contemporaneous HICP level, at 0.618, while its correlation with the preceding quarterly change is considerably smaller at 0.268. The model measure is less strongly related to the level, at 0.229, and more strongly related to the preceding quarterly change, at 0.338. This pattern is consistent with the model measure being less dominated by contemporaneous inflation levels and more responsive to directional changes in inflation than the dictionary measure. It does not, however, establish that the model successfully measures a predictive inflation outlook: neither measure is meaningfully

associated with the change in HICP over the following two quarters, with correlations of -0.011 for the model and -0.041 for the dictionary.

8.5 Comparison between the Contemporaneous Correlations

The difference between the two correlations with the HICP level can be tested directly rather than inferred from their magnitudes. The two outlook measures are themselves correlated at 0.482 and are evaluated against the same HICP series over the same quarters, so the two estimated correlations are not independent.

Williams’s test provides a conventional test for dependent correlations that share a common variable and gives a statistic of 4.93 on 105 degrees of freedom. Its standard implementation, however, assumes independent observations, which is inappropriate for the quarterly time series considered here. The primary comparison therefore uses an autocorrelation-robust formulation. Writing z for a standardized variable, define

$$d_t = z_t^{\text{HICP}} z_t^{\text{dict}} - z_t^{\text{HICP}} z_t^{\text{model}}.$$

The mean of d_t is the difference between the two sample correlations, $r_{\text{dict}} - r_{\text{model}}$. Testing whether this mean equals zero therefore provides a direct test of whether the dictionary and model have the same relationship with contemporaneous HICP. A Newey–West standard error is applied to the mean of d_t to account for serial dependence in the quarterly observations.

Table 16: Difference in correlation with the HICP level

Test	Difference	Std. error	t	p
Williams, $n = 108$	+0.388	—	+4.93	< 0.0001
Newey–West on d_t	+0.388	0.163	+2.38	0.019

Newey–West with four lags; 95% interval [+0.069, +0.708]. A single pre-committed test, so no multiplicity adjustment applies.

The estimated difference in correlations is +0.388. Allowing for serial dependence reduces the strength of the conventional Williams result, but the difference remains statistically distinguishable from zero: the Newey–West standard error is 0.163, giving $t = 2.38$, $p = 0.019$, and a 95% interval of [+0.069, +0.708]. The two measures therefore have significantly different associations with contemporaneous inflation, despite being constructed from identical text and intended to represent the same broad inflation-outlook concept.

8.6 Interpretation and Limitations

The stronger contemporaneous HICP correlation of the dictionary should not by itself be interpreted as evidence that it provides the better inflation-outlook measure. The intended construct distinguishes the direction of the inflation outlook from the current inflation level. Lexical expressions associated with elevated inflation or price pressures can occur in descriptions of both current conditions and expected developments, making it difficult for a keyword-based procedure to distinguish the two without richer contextual information. The results are consistent with this limitation: the dictionary measure is substantially more closely associated with the contemporaneous inflation level, while the model measure is relatively more associated with recent directional changes.

The broader methodological implication is that the quality of a constructed text measure cannot be assessed solely by the magnitude of its correlation with an external variable. The relevant external relationship depends on the construct the measure is intended to represent. A high correlation may indicate strong criterion validity with respect to a particular external series while simultaneously revealing sensitivity to a dimension that the coding scheme was intended to separate. External validation therefore needs to be interpreted jointly with the definition of the underlying construct rather than as a comparison in which the largest correlation necessarily identifies the superior measurement procedure.

Neither measure is meaningfully associated with subsequent realized inflation: the correlations with the two-quarter-ahead HICP change are -0.011 for the model and -0.041 for the dictionary. This result clarifies the interpretation of the outlook measure. “Forward-looking” refers to the direction of inflation expectations expressed in the speech, rather than to the ex post accuracy of those expectations. The measure should therefore be interpreted as capturing the communicated inflation outlook, not as a predictor of subsequently realized inflation.

Three limitations qualify the benchmark. First, the dictionary is constructed specifically for this project rather than taken from an established published instrument. The comparison therefore shows how a lexical implementation of this coding scheme performs relative to the LLM classification; it does not establish that no dictionary-based method could distinguish current conditions from directional outlook. Second, the dictionary thresholds are calibrated using a corpus that includes the thirty audit speeches. This introduces a small degree of in-sample information into the benchmark, although those speeches constitute only thirty of the 2,717 observations. Third, both methods are evaluated against the single non-blind human coder from the initial validation round. The agreement rates should therefore be interpreted as comparative performance against the same reference classifications rather than as definitive estimates of either method’s accuracy.

9 Conclusion

This project develops and evaluates a pipeline for constructing economic variables from central-bank communication using a large language model. Its purpose is methodological and technical rather than to provide a finished empirical analysis of ECB communication. The exercise begins with speech-level classifications, subjects them to successive rounds of human validation, aggregates them into economically interpretable communication measures, compares those measures with external macroeconomic variables, and benchmarks the extraction against a simpler dictionary-based method. The resulting analysis illustrates both what can be learned from an LLM-based measurement procedure and what must remain unresolved when the validation evidence is incomplete.

The first question was whether the model reproduces human coding of the same text and how disagreements should be interpreted. The validation sequence shows why this cannot be answered from agreement with a single human coder alone. Initial model–human agreement was high, but the introduction of a second independent coder revealed substantial variation in the reproducibility of the coding task itself. Human–human agreement was 0.694 on the round-three identical-text sample, with particularly weak agreement for several individual variables. Inspection of these disagreements subsequently identified both discrepancies between the human codebook and the production prompt and decision boundaries that required further clarification. After the coding scheme was revised, the same two coders independently classified fifty previously unseen speeches under the new rules and agreed on 0.810 of classifications. Because the samples and sampling designs differ, this increase cannot be attributed exclusively to the codebook revision, but it provides descriptive evidence of greater reproducibility under the revised procedure. More generally, the exercise shows that model–human disagreement cannot automatically be interpreted as model error when the human reference classification is itself uncertain.

The comparison with the dictionary benchmark provides a complementary answer to what the language model contributes. The advantage of contextual classification is not uniform across tasks. For inflation attention, the model and dictionary perform identically against the initial human reference, suggesting that lexical information is sufficient to capture much of this relatively simple distinction. Larger differences emerge where classification depends on context rather than vocabulary alone: the dictionary generates substantially more false positives for growth attention and misses uncertainty classifications that require distinguishing substantive uncertainty from routine references to risk. Across all 180 classifications in the initial audit, model–human agreement is 0.944, compared with 0.728 for the dictionary. The inflation-outlook comparison further shows that agreement or external correlation cannot be interpreted independently of the construct being measured. The dictionary correlates much more strongly with the contemporaneous HICP level than the model, at 0.618 against 0.229, while the model is relatively more associated with recent directional changes in inflation. The two methods therefore extract different information from the same text, despite being designed to represent the same broad concept.

The constructed communication measures also provide some evidence that the distinctions imposed by the measurement framework have empirical content. Attention and conditional outlook capture different dimensions of communication: in the growth application they are only weakly correlated with one another and exhibit significantly different relationships with the same subsequent unemployment target. More broadly, all five pre-specified external-validation relationships have the expected direction, although none survives adjustment for multiple testing. These results are therefore best interpreted as evidence that the measures behave broadly in accordance with their intended economic interpretation rather than as established relationships between ECB communication and macroeconomic outcomes. They provide neither causal evidence nor evidence of forecasting performance. The misclassification exercise leads to a similar distinction between what can be implemented and what can currently be estimated credibly: the correction for binary classification error is straightforward once sensitivity and specificity are known, but the successive validation rounds show that the available error-rate estimates are not yet sufficiently stable to treat the resulting corrected series as preferred estimates.

Several steps therefore remain before the measurement framework could support a finished empirical application. Most immediately, the production extraction should be repeated under the complete revised coding specification and evaluated on the round-four speeches, allowing the model and human coders to be compared under the same rules and on the same observations. That validation could then provide a stronger basis for estimating the misclassification parameters used in the aggregate correction. The uncertainty construct requires further examination, since inter-coder reliability remains weak after targeted clarification; a larger validation sample and additional independent coders would provide a stronger estimate of human reproducibility; and systematic prompt-sensitivity analysis would establish whether the extracted measures are robust to reasonable changes in the model specification. A subsequent empirical application should also propagate uncertainty from the measurement stage into downstream inference. These omissions are material, but they also define the next steps of the project. The current exercise therefore stops short of presenting the extracted series as finished economic measurements and instead provides a reproducible framework for identifying where an LLM-based measurement pipeline succeeds, where its validation fails, and what evidence would be required before extending it further.

References

- Blinder, A. S., Ehrmann, M., Fratzscher, M., De Haan, J., and Jansen, D.-J. (2008). Central bank communication and monetary policy: A survey of theory and evidence. *Journal of economic literature*, 46(4):910–945.
- Clopper, C. J. and Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413.
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and psychological measurement*, 20(1):37–46.
- Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–574.
- Hansen, S. and McMahon, M. (2016). Shocking language: Understanding the macroeconomic effects of central bank communication. *Journal of International Economics*, 99:S114–S133.
- Hausman, J. A., Abrevaya, J., and Scott-Morton, F. M. (1998). Misclassification of the dependent variable in a discrete-response setting. *Journal of econometrics*, 87(2):239–269.
- Hollo, D., Kremer, M., and Lo Duca, M. (2012). Ciss-a composite indicator of systemic stress in the financial system.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian journal of statistics*, pages 65–70.
- Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The annals of Statistics*, pages 1217–1241.
- Lahiri, K. and Sheng, X. (2010). Measuring forecast uncertainty by disagreement: The missing link. *Journal of Applied Econometrics*, 25(4):514–538.
- Loughran, T. and McDonald, B. (2011). When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of finance*, 66(1):35–65.
- Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., and Yang, D. (2024). Can large language models transform computational social science? *Computational linguistics*, 50(1):237–291.

Part II: Statistical and Methodological Appendix

This appendix provides the statistical derivations and methodological details needed to interpret the empirical procedures used in Part I. Its organization follows the sequence of the analysis rather than providing a general treatment of the underlying econometric methods.

The first section concerns measurement. It defines the agreement and classification statistics used in the human-validation exercises, shows how sensitivity and specificity are recovered under the stratified sampling design of round two, and explains the exact intervals used for validation proportions. It then derives the binary misclassification correction used illustratively in Section 5 and shows how uncertainty in the estimated error rates is propagated into corrected attention shares.

The second section records the application-specific choices needed for time-series inference. Newey–West HAC inference itself is standard; the relevant issue here is the choice of bandwidth, particularly because the two-quarter unemployment target overlaps across adjacent observations. The section therefore documents the bandwidth rule, its sensitivity, and the lower bound implied by the overlapping outcome.

The remaining sections contain two calculations that are less immediate from the reported results. The first shows how differences between two correlations sharing the same target are tested while preserving their sampling dependence. The second documents the construction of the dictionary benchmark, including its normalized attention score, directional measure, prevalence-matched threshold, and neutral-band calibration.

The purpose of the appendix is therefore limited to calculations that are specific to the empirical design or are not transparent from the reported statistics alone. Standard procedures used without modification are stated in Part I rather than re-derived here.

A Validation Statistics and Misclassification

A.1 Agreement and Classification Rates

Let $H \in \{0, 1\}$ denote a human classification and $M \in \{0, 1\}$ the corresponding model classification. Raw agreement is the proportion of observations for which $H = M$. Because agreement also depends on the marginal frequencies with which the available categories are used, Section 4 reports Cohen’s κ alongside raw agreement.

For two coders assigning categories from a finite set $\mathcal{C}$, let p_o denote observed agreement and let

$$p_e = \sum_{c \in \mathcal{C}} \Pr(H = c) \Pr(M = c)$$

denote the agreement expected if the two coders classified observations independently while retaining their observed marginal category frequencies. In the binary case this reduces to

$$p_e = \Pr(H = 1) \Pr(M = 1) + \Pr(H = 0) \Pr(M = 0).$$

Cohen’s statistic is (Cohen, 1960)

$$\kappa = \frac{p_o - p_e}{1 - p_e}. \quad (10)$$

Thus $\kappa = 1$ under perfect agreement and $\kappa = 0$ when observed agreement equals the agreement implied by independent codings with the same marginals. Because κ can vary substantially with those marginals, neither it nor raw agreement is treated as a sufficient summary on its own.

For the binary attention measures, the joint classifications can be summarized by the 2×2 confusion matrix

	$H = 1$	$H = 0$
$M = 1$	TP	FP
$M = 0$	FN	TN

where true positives (TP) and true negatives (TN) denote cases in which the model and human reference agree on a positive or negative classification, respectively, while false positives (FP) and false negatives (FN) denote the two possible directions of disagreement. These labels follow the conventional confusion-matrix terminology; the human classification is treated as a reference rather than as an independently observed ground truth.

Taking the human classification as the reference,

$$\alpha = \Pr(M = 1 \mid H = 1), \quad \beta = \Pr(M = 0 \mid H = 0)$$

are sensitivity and specificity, with sample analogues

$$\hat{\alpha} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \hat{\beta} = \frac{\text{TN}}{\text{TN} + \text{FP}}.$$

Sensitivity measures the share of reference-positive cases also coded positive by the model; specificity measures the share of reference-negative cases also coded negative. Reporting the two separately matters because the observed classification disagreements are asymmetric.

These rates measure model performance against the designated human classification, not accuracy against an independently observed latent truth. The human–human comparisons in Sections 4.3–4.6 show why this distinction matters: where independent human coders disagree, model–human disagreement cannot by itself be attributed to model error.

A.2 Recovery Under Choice-Based Validation Sampling

Round two does not use a simple random validation sample. Sampling is stratified partly on the classifications produced by the model, with model-positive speeches deliberately oversampled. Consequently, raw validation-sample sensitivity and specificity do not generally estimate their counterparts in the baseline corpus.

Let $S = 1$ indicate that a speech is selected into the validation sample. The sampling probabilities are determined by the model’s classification M , not by the human classification H . Thus, among speeches with the same model classification, selection does not depend on how the human coder would classify the speech. Formally, the required assumption is

$$S \perp H \mid M. \tag{11}$$

Under (11), quantities that condition on M are preserved by the sampling design:

$$\Pr(H = 1 \mid M = 1, S = 1) = \Pr(H = 1 \mid M = 1) \equiv \text{PPV},$$

and

$$\Pr(H = 0 \mid M = 0, S = 1) = \Pr(H = 0 \mid M = 0) \equiv \text{NPV}.$$

Oversampling model-positive speeches changes how frequently the $M = 1$ stratum appears in the validation sample, but not the proportion of human-positive observations within that stratum, provided (11) holds. The same reasoning applies to NPV within the $M = 0$ stratum.

Sensitivity and specificity instead condition on the human classification H . Because the stratified design changes the proportion of model-positive and model-negative speeches in the validation sample, their raw validation-sample analogues are generally distorted. The corpus-level proportion of the two model classifications is, however, observed because the model code is available for every speech. Let

$$q = \Pr(M = 1)$$

denote the model-positive rate in the full baseline corpus. Combining q with the validation-sample estimates of PPV and NPV reconstructs the joint distribution of H and M in the baseline corpus.

In particular,

$$\Pr(H = 1, M = 1) = \text{PPV} \cdot q,$$

while

$$\Pr(H = 1, M = 0) = (1 - \text{NPV})(1 - q).$$

Letting

$$p = \Pr(H = 1),$$

the law of total probability therefore gives

$$\begin{aligned} p &= \Pr(H = 1, M = 1) + \Pr(H = 1, M = 0) \\ &= \text{PPV} \cdot q + (1 - \text{NPV})(1 - q). \end{aligned} \quad (12)$$

Bayes's rule then gives

$$\alpha = \Pr(M = 1 \mid H = 1) = \frac{\text{PPV} \cdot q}{p}, \quad (13)$$

$$\beta = \Pr(M = 0 \mid H = 0) = \frac{\text{NPV}(1 - q)}{1 - p}. \quad (14)$$

The model-positive rate q is computed directly from the full baseline corpus, for which model classifications are available for every speech. By contrast, PPV and NPV are estimated from the smaller human-coded validation sample and are therefore subject to sampling uncertainty. Thus, conditional on the observed corpus-level value of q , uncertainty in the recovered sensitivity and specificity arises from the estimation of PPV and NPV.

When $\text{NPV} = 1$, as observed for all four attention variables in round two,

$$p = \text{PPV} \cdot q,$$

and hence

$$\alpha = \frac{\text{PPV} \cdot q}{\text{PPV} \cdot q} = 1. \quad (15)$$

The four reported sensitivity estimates of 1.000 therefore reflect the same underlying sample feature—the absence of observed false negatives—rather than four independent pieces of evidence for perfect sensitivity.

The recovery in (12)–(14) depends on (11). If selection also depended on characteristics related to H after conditioning on M , PPV and NPV estimated in the validation sample would no longer identify their corpus counterparts.

A.3 Exact Intervals for Validation Proportions

Several validation rates are estimated from small denominators, and some are observed at the boundary. The analysis therefore uses exact Clopper–Pearson intervals for binomial proportions (Clopper and Pearson, 1934).

For k successes in n Bernoulli trials, a two-sided $100(1 - \delta)\%$ interval is

$$p_L = B^{-1}\left(\frac{\delta}{2}; k, n - k + 1\right), \quad p_U = B^{-1}\left(1 - \frac{\delta}{2}; k + 1, n - k\right),$$

with $p_L = 0$ when $k = 0$ and $p_U = 1$ when $k = n$.

The use of an exact interval matters at the boundary. If ten sampled observations are all successes, an ordinary nonparametric bootstrap resamples only successes and therefore returns the degenerate interval $[1, 1]$. The exact 95% binomial interval instead has lower endpoint 0.6915. A sample estimate of 1.000 therefore does not imply that the corresponding population rate is known to equal one.

Confidence intervals for Cohen's κ are treated separately, because κ is not a binomial proportion. Those intervals are obtained by a percentile bootstrap in which the resampling unit is the speech.

A.4 Misclassification Correction

Section 5 illustrates how estimated classification error can alter an aggregate attention share. Let $D^* \in \{0, 1\}$ denote the reference classification and $D \in \{0, 1\}$ the classification produced by the instrument. Define

$$\alpha = \Pr(D = 1 \mid D^* = 1), \quad \beta = \Pr(D = 0 \mid D^* = 0),$$

and let

$$p = \Pr(D^* = 1), \quad q = \Pr(D = 1).$$

Under a stable non-differential misclassification process, the observed positive share consists of correctly classified reference positives and incorrectly classified reference negatives:

$$\begin{aligned} q &= \Pr(D = 1 \mid D^* = 1) \Pr(D^* = 1) + \Pr(D = 1 \mid D^* = 0) \Pr(D^* = 0) \\ &= \alpha p + (1 - \beta)(1 - p) \\ &= (1 - \beta) + p(\alpha + \beta - 1). \end{aligned} \tag{16}$$

Writing

$$J = \alpha + \beta - 1,$$

the relationship can be inverted whenever $J \neq 0$:

$$p = \frac{q - (1 - \beta)}{J}. \tag{17}$$

The implementation requires $J > 0$, corresponding to a classifier whose codes are positively informative about the reference classification. If $J = 0$, then

$$\Pr(D = 1 \mid D^* = 1) = \Pr(D = 1 \mid D^* = 0),$$

so the probability of a positive instrument classification does not depend on the reference state and the observed positive share contains no information with which to recover p .

Replacing q , α and β by the corpus share and validation estimates gives the correction used in Section 5. The exercise is illustrative rather than a preferred estimate of the underlying attention series: subsequent validation shows that the estimated error rates depend on the validation design and on the human classification used as the reference.

The transformation is not generally a constant level shift. In the special case $\alpha = 1$, as estimated for the attention variables in the initial audit,

$$\hat{p} = \frac{q - (1 - \beta)}{\beta},$$

and therefore

$$\begin{aligned} q - \hat{p} &= q - \frac{q - 1 + \beta}{\beta} \\ &= \frac{(1 - \beta)(1 - q)}{\beta}. \end{aligned} \tag{18}$$

The correction therefore varies with the observed share. When false positives are the only estimated classification error, the adjustment is larger when observed attention is lower because more observations remain available to be incorrectly classified as positive. This is the feature illustrated in Figure 2.

A.5 Uncertainty in Corrected Shares

The correction in (17) depends on estimated sensitivity and specificity, so uncertainty in those rates must also be propagated into the corrected share.

For fixed q ,

$$\hat{p}(\alpha, \beta) = \frac{q - 1 + \beta}{\alpha + \beta - 1}.$$

The implementation evaluates this expression at all four combinations of the endpoints of

$$[\alpha_L, \alpha_U] \times [\beta_L, \beta_U],$$

retaining combinations for which

$$\alpha + \beta - 1 > 0,$$

and reports the minimum and maximum corrected values. The informativeness restriction does not bind for the validation intervals used in Section 5; at the tightest evaluated corner, $\alpha = 0.692$ and $\beta = 0.621$, giving $J = 0.313$.

Corrected shares are also constrained to the admissible interval $[0, 1]$ before the extrema are taken. A reported endpoint can therefore reflect this logical bound rather than an unconstrained value attained exactly at a confidence-box corner. In particular, the upper bound of 1.000 reported for growth attention results from this clipping step.

These ranges propagate uncertainty in the estimated sensitivity and specificity intervals. They are not a complete confidence region for the corrected series, because they do not incorporate uncertainty about whether a common misclassification process applies across periods or other speech characteristics.

A.6 Dependence Between Attention and Outlook Classifications

The coding scheme imposes the logical restriction

$$A_i^x = 0 \implies O_i^x = 0.$$

An outlook disagreement can therefore arise mechanically when coders disagree about the corresponding attention indicator. Such a case does not provide independent evidence that the coders interpreted the direction of the outlook differently. Section 4 consequently distinguishes disagreements induced by the attention classification from those occurring when both coders agree that the topic is substantively discussed.

The restriction also creates dependence between classification cells within a speech. Pooled cell-level agreement therefore does not consist of independent Bernoulli observations. For the fifty-speech round-four sample, human–human agreement is 0.810, with the reported binomial 95% interval $[0.761, 0.853]$ and a speech-cluster bootstrap interval of $[0.747, 0.867]$. The corresponding model–human rates are 0.737, with intervals $[0.683, 0.786]$ and $[0.677, 0.793]$, and 0.743, with intervals $[0.690, 0.792]$ and $[0.690, 0.797]$, respectively.

Allowing for within-speech dependence therefore widens the uncertainty without changing the descriptive ordering of the point estimates. The ordering should nevertheless not be interpreted as a ranking of accuracy: human–human agreement measures reproducibility under the coding scheme, while model–human agreement measures correspondence with a fallible human reference.

B HAC Bandwidth and Overlapping Outcomes

The external-validation correlations in Section 7 use Newey–West heteroskedasticity- and autocorrelation-consistent standard errors (?). The estimator itself is standard; the application-specific choice is the bandwidth. The analysis uses

$$L = \max \left\{ \left\lfloor 4(T/100)^{2/9} \right\rfloor, h - 1 \right\}, \quad (19)$$

where T is the number of quarterly observations and h is the horizon entering the construction of the dependent variable.

The lower bound matters for the two-quarter unemployment change,

$$y_t^{(h)} = u_{t+h} - u_t, \quad h = 2.$$

Adjacent observations overlap. Under the illustrative process

$$u_{t+1} = u_t + \eta_{t+1},$$

with serially independent innovations,

$$y_t^{(h)} = \sum_{j=1}^h \eta_{t+j},$$

and

$$\text{Cov}\left(y_t^{(h)}, y_{t-\ell}^{(h)}\right) = (h - \ell)\sigma_\eta^2 \quad \text{for } 0 \leq \ell < h. \quad (20)$$

An overlapping h -period change therefore contains an $\text{MA}(h - 1)$ component even when the underlying innovations are serially independent. With $h = 2$, the lower bound in (19) is one.

For integer sample sizes in the range relevant here, the automatic rule gives

$$L = 3 \quad \text{for } 28 \leq T \leq 99, \quad L = 4 \quad \text{for } 100 \leq T \leq 272.$$

The five principal specifications contain between 102 and 108 quarters, so $L = 4$ throughout and the overlap-based floor does not bind.

Because the samples lie close to the threshold between bandwidths three and four, inference is recomputed at $L = 3$ and $L = 6$. Point estimates are unchanged because the bandwidth affects the covariance estimator rather than the correlations themselves. Across the five pre-specified relationships, the largest movement in any raw p -value is 0.045. Growth outlook against the two-quarter unemployment change remains individually significant throughout, with $p = 0.017$, 0.018 and 0.021 at $L = 3$, 4 and 6, respectively. The other four relationships remain above the 5% threshold at all three bandwidths, and none of the five relationships survives the Holm family-wise adjustment at any of them. The interpretation of the external-validation exercise is therefore not sensitive to the bandwidth selected by the automatic rule.

C Dictionary Benchmark Construction

The benchmark in Section 8 compares the LLM classification with a transparent rule-based alternative applied to the same text. Because the dictionary produces continuous scores while the coding scheme requires discrete classifications, the thresholds used to convert those scores into classifications form part of the measurement procedure.

C.1 Attention as a Normalized Density

For topic x and speech i containing W_i words, let M_i^x denote the number of matches to the topic vocabulary. Dictionary attention is measured as

$$d_i^x = 1000 \frac{M_i^x}{W_i}.$$

Using a match density rather than a raw count reduces the mechanical relationship between the dictionary score and speech length, since a longer speech otherwise has more opportunities to contain vocabulary matches.

C.2 Direction within Topic Sentences

Directional language is scored only within sentences containing the relevant topic vocabulary. Let S_i^x denote the set of topic sentences, and let $U(s)$ and $D(s)$ count upward and downward directional terms in sentence s . Define

$$\text{tone}_i^x = \frac{1}{|S_i^x|} \sum_{s \in S_i^x} [U(s) - D(s)],$$

with tone equal to zero when no topic sentence is present.

A directional term is treated as negated when a negator occurs within the preceding three words. A negated upward term contributes in the downward direction, and conversely. The three-word window is an implementation choice; sensitivity to alternative window lengths is not examined. Restricting the calculation to topic sentences prevents directional language about unrelated subjects from determining the topic-specific outlook classification.

C.3 Prevalence-Matched Thresholds

The attention density is continuous, whereas the benchmark must produce a binary classification. Rather than choosing the cutoff from validation performance, the threshold is calibrated to approximately reproduce the model’s corpus-level positive rate.

Let π denote the share of baseline speeches for which the model assigns attention equal to one. The dictionary threshold is

$$c = \widehat{Q}_d(1 - \pi), \quad \widehat{\Pr}(d > c) \approx \pi.$$

The comparison therefore focuses primarily on which speeches the two methods classify as positive rather than allowing differences in their overall positive rates to be determined by an arbitrary threshold. Exact matching need not occur when the density distribution contains ties, which is why Section 8 reports the realised classification rates.

The calibration gives the dictionary information about the model’s prevalence that an independently deployed dictionary would not ordinarily possess. The benchmark is therefore deliberately favourable to the simpler procedure with respect to this dimension.

C.4 Neutral-Band Calibration

The ternary outlook classification requires an additional threshold around zero. Because outlook is mechanically zero whenever dictionary attention is zero, the neutral-band calibration must first account for those forced zeros.

Let a denote the share of speeches for which the dictionary assigns attention equal to one, and let z denote the share of zero outlook classifications produced by the model. A share $1 - a$ of observations is already forced to zero by the attention restriction. The additional neutral share required among attending speeches is therefore

$$z - (1 - a).$$

Expressed relative to the attending subset,

$$w = \frac{z - (1 - a)}{a}. \tag{21}$$

The neutral band is set at the w -quantile of $|\text{tone}|$ among attending speeches. Attending speeches inside the band are coded zero, while those outside it are assigned the sign of their directional score. When

$$z - (1 - a) \leq 0,$$

the attention restriction alone already generates at least the target share of neutral classifications, so the band is set to zero.

Restricting the quantile calculation to attending speeches prevents the forced zeros from entering the calibration twice.

Part III: Measurement and Replication Materials

D Reproducing the Results

Every number in Part I is produced by a script in the repository. The run order is given in Table 17; each script writes to `outputs/` and prints its principal results to standard output.

Table 17: Pipeline and principal outputs

Stage	Script	Principal output
Corpus download	<code>src/download_data.py</code>	raw ECB speeches file
Corpus audit	<code>src/audit_data.py</code>	corpus counts and coverage
Cleaning	<code>src/clean_text.py</code>	baseline sample
Excerpts	<code>src/prepare_full_sample.py</code>	2,000-word excerpts
Extraction	<code>src/extract_signals_openai.py</code>	six signals (do not re-run)
Completeness	<code>src/validate_extractions.py</code>	coverage checks
Speech panel	<code>analysis/10_panel.py</code>	<code>speech_panel.csv</code> ; Table 1
Round-one audit	<code>analysis/15_validation_stats.py</code>	Table 9; error-rate JSON
Aggregation	<code>analysis/12_measures.py</code>	monthly/quarterly/annual measures; Table 10; Figure 2
Macro download	<code>src/download_macro.py</code>	<code>macro_quarterly.csv</code>
Salience download	<code>src/download_salience_series.py</code>	<code>salience_quarterly.csv</code>
External concordance	<code>analysis/13_external_validation.py</code>	<code>external_concordance.csv</code>
Dictionary benchmark	<code>analysis/16_dictionary_benchmark.py</code>	Tables 15 and 16; Figure 4
Validation draw	<code>src/select_validation_sample.py</code>	seeded blind sample and key
Coding sheets	<code>src/build_coding_sheet.py</code>	blind sheets and speech texts
Round two	<code>analysis/17_validation_round2.py</code>	Table 3
Round three	<code>analysis/18_coder_reliability.py</code>	Tables 4, 5 and 6
Specification diff	<code>src/diff_prompts.py</code>	version 1 to version 2 rule diff
Round four	<code>analysis/19_round4_ceiling.py</code>	Tables 7 and 8; Figure 1
External validation	<code>analysis/25_salience_validation.py</code>	Tables 11, 12, 13 and 14; Figure 3

The draw and coding-sheet stages are run once per validation round, with `ACTIVE_ROUND` set to the round being drawn; the seeded keys for rounds two and four are committed, so the analysis stages that consume them reproduce without redrawing. The human coding files in `data/human/` are inputs to the validation stages rather than outputs of them.

Four supporting modules contain the reusable logic. `src/measurement_error.py` implements the binary misclassification correction, `src/agreement.py` contains the agreement statistics used in the audit and benchmark, `src/timeseries_stats.py` contains the time-series inference procedures, and `src/dictionary_signals.py` contains the benchmark vocabularies and scoring rules.

The codebase also contains explicit consistency checks for quantities whose silent failure could otherwise produce plausible but incorrect results. The aggregation checks the relationship between attention-weighted and conditional outlook measures; the external-validation routines check the quarterly index and required sample coverage; the audit checks agreement flags against the recorded codes; and the correlation-difference routines verify the relevant algebraic identities. The bandwidth-sensitivity routine additionally derives the sample-size ranges associated with the automatic Newey–West rule and verifies their boundaries against the implemented bandwidth function, so that a subsequent change in the underlying rule causes the check to fail rather than leaving the documented ranges silently unchanged.

The original extraction is treated as fixed throughout the analysis and is not regenerated as part of the replication pipeline. This distinction is important for the material below: the production classification prompt generated the corpus used in Part I, whereas the proposed prompt incorporates

decision rules developed only after the subsequent validation and adjudication exercises.

E Production LLM Extraction Prompt

The following prompt is the specification used for the production LLM-based extraction reported in Part I. It is reproduced from `prompts/extraction_prompt.txt`. The placeholders `{title}`, `{subtitle}`, and `{text}` are replaced by the corresponding speech-level inputs at inference time. The doubled braces in the required-output block are intentional escapes in the `str.format()` template and render as single braces in the prompt supplied to the model.

```
You are extracting structured variables from European Central Bank communication for
economic research.

Classify the supplied ECB speech text using only the information contained in the supplied
text. Do not use outside knowledge about the speaker, date, economic conditions,
monetary-policy decisions, or subsequent events.

The objective is measurement, not summarization.

Return the following six variables:

1. inflation_attention
    * 1: inflation, price developments, price pressures, inflation expectations, or closely
      related inflation developments are substantively discussed.
    * 0: they are not substantively discussed.
    * A passing reference to inflation or price stability is not sufficient.
2. inflation_outlook
    * -1: the dominant forward-looking emphasis is disinflationary or points to downside
      inflation developments or risks.
    * 0: the discussion is balanced, neutral, mixed, ambiguous, purely descriptive, or
      inflation is not substantively discussed.
    * 1: the dominant forward-looking emphasis is inflationary or points to upside inflation
      developments or risks.
    * Classify the direction of the forward-looking inflation outlook or dominant expected
      trajectory, not merely whether the current level of inflation is high or low.
    * If inflation_attention = 0, then inflation_outlook must equal 0.
3. growth_attention
    * 1: real economic activity, including GDP, output, demand, consumption, investment,
      recession, recovery, or closely related real-activity developments, is substantively
      discussed.
    * 0: real economic activity is not substantively discussed.
    * A passing reference is not sufficient.
4. growth_outlook
    * -1: the dominant forward-looking emphasis is weakening activity, deterioration,
      recessionary conditions, or downside growth risks.
    * 0: the discussion is balanced, neutral, mixed, ambiguous, purely descriptive, or
      growth is not substantively discussed.
    * 1: the dominant forward-looking emphasis is strengthening activity, recovery,
      improving conditions, or upside growth developments.
    * Classify the direction of the forward-looking growth outlook or dominant expected
      trajectory, not merely whether the current level of economic activity is strong or weak.
    * If growth_attention = 0, then growth_outlook must equal 0.
5. financial_stability_attention
    * 1: financial-system risks or resilience are substantively discussed, including banking
      -system vulnerabilities, solvency or liquidity stress, systemic risk, contagion, market
      stress, macroprudential risks, or financial fragility.
    * 0: these issues are not substantively discussed.
    * Do not classify ordinary references to financial markets, interest rates, or monetary-
      policy transmission as financial-stability discussion unless systemic or stability-
      related concerns are substantively addressed.
6. 'uncertainty_attention'
```

- '1': uncertainty, unpredictability, unusually wide risks, alternative scenarios, or uncertainty surrounding the economic outlook are substantively emphasized.
- '0': they are not substantively emphasized.
- The mere occurrence of words such as "risk" or routine statements such as "risks are broadly balanced" is not sufficient.
- Do not code '1' merely because technical, operational, legal, credit, settlement, or financial-system risks are discussed. Such risks count as uncertainty attention only when the text substantively emphasizes uncertainty, unpredictability, unusually wide possible outcomes, or difficulty assessing the outlook.

General rules:

- * Base every classification only on the supplied text.
- * Classify substantive economic discussion rather than keyword presence.
- * Do not infer views that are not supported by the text.
- * For outlook variables, classify the dominant expected direction rather than counting positive and negative words.
- * Distinguish current conditions from the forward-looking outlook. For example, weak current activity can coexist with a positive growth outlook if the text clearly emphasizes an expected recovery.
- * When forward-looking evidence is genuinely mixed or insufficient, use 0.
- * Do not summarize the speech.
- * Do not add variables, explanations, commentary, confidence scores, or additional text.
- * Return only the required structured output.

Required output structure:

```
{{
"inflation_attention": 0,
"inflation_outlook": 0,
"growth_attention": 0,
"growth_outlook": 0,
"financial_stability_attention": 0,
"uncertainty_attention": 0
}}
```

Speech metadata and text follow below.

TITLE:
{title}

SUBTITLE:
{subtitle}

TEXT:
{text}

F Revised LLM Extraction Prompt

The following prompt is the proposed version developed after the validation and adjudication exercises. It is reproduced from `prompts/extraction_prompt_v4.txt` and was not used to generate the production extraction analysed in Part I.

The distinction between the production and revised specifications is important. Three repairs described in Table 18 correct discrepancies in the human codebook by restoring rules that were already present in the production prompt. They therefore do not represent changes from the production prompt to the new version. The revision instead adds six decision rules developed from ambiguities identified during adjudication: conflicting inflation horizons; payment and operational resilience; the threshold for uncertainty attention; the threshold for substantive attention; attention without a forecast; and the distinction between an outlook and a hypothetical or causal mechanism.

As in the production template, the placeholders {title}, {subtitle}, and {text} are replaced at inference time, and the doubled braces in the output block are intentional template escapes.

You are extracting structured variables from European Central Bank communication for economic research.

Classify the supplied ECB speech text using only the information contained in the supplied text. Do not use outside knowledge about the speaker, date, economic conditions, monetary-policy decisions, or subsequent events.

The objective is measurement, not summarization.

Return the following six variables:

1. inflation_attention
 - * 1: inflation, price developments, price pressures, inflation expectations, or closely related inflation developments are substantively discussed.
 - * 0: they are not substantively discussed.
 - * A passing reference to inflation or price stability is not sufficient.
2. inflation_outlook
 - * -1: the dominant forward-looking emphasis is disinflationary or points to downside inflation developments or risks.
 - * 0: the discussion is balanced, neutral, mixed, ambiguous, purely descriptive, or inflation is not substantively discussed.
 - * 1: the dominant forward-looking emphasis is inflationary or points to upside inflation developments or risks.
 - * Classify the direction of the forward-looking inflation outlook or dominant expected trajectory, not merely whether the current level of inflation is high or low.
 - * When near-term headline inflation and an explicitly stated medium-term or policy-relevant inflation outlook point in opposite directions, classify using the medium-term policy-relevant horizon. A transitory near-term movement does not overturn an explicit medium-term direction.
 - * If inflation_attention = 0, then inflation_outlook must equal 0.
3. growth_attention
 - * 1: real economic activity, including GDP, output, demand, consumption, investment, recession, recovery, or closely related real-activity developments, is substantively discussed.
 - * 0: real economic activity is not substantively discussed.
 - * A passing reference is not sufficient.
4. growth_outlook
 - * -1: the dominant forward-looking emphasis is weakening activity, deterioration, recessionary conditions, or downside growth risks.
 - * 0: the discussion is balanced, neutral, mixed, ambiguous, purely descriptive, or growth is not substantively discussed.
 - * 1: the dominant forward-looking emphasis is strengthening activity, recovery, improving conditions, or upside growth developments.
 - * Classify the direction of the forward-looking growth outlook or dominant expected trajectory, not merely whether the current level of economic activity is strong or weak.
 - * If growth_attention = 0, then growth_outlook must equal 0.
5. financial_stability_attention
 - * 1: financial-system risks or resilience are substantively discussed, including banking-system vulnerabilities, solvency or liquidity stress, systemic risk, contagion, market stress, macroprudential risks, or financial fragility.
 - * 0: these issues are not substantively discussed.
 - * Do not classify ordinary references to financial markets, interest rates, or monetary-policy transmission as financial-stability discussion unless systemic or stability-related concerns are substantively addressed.
 - * Safety, security, settlement, payment-system continuity or operational resilience does not by itself constitute financial-stability attention. Code 1 only when the discussion is explicitly tied to system-wide impairment, contagion, banking-system resilience, macroprudential risk, or the capacity of financial intermediation to serve the economy.
6. 'uncertainty_attention'

- '1': uncertainty, unpredictability, unusually wide risks, alternative scenarios, or uncertainty surrounding the economic outlook are substantively emphasized.
- '0': they are not substantively emphasized.
- The mere occurrence of words such as "risk" or routine statements such as "risks are broadly balanced" is not sufficient.
- Do not code '1' merely because technical, operational, legal, credit, settlement, or financial-system risks are discussed. Such risks count as uncertainty attention only when the text substantively emphasizes uncertainty, unpredictability, unusually wide possible outcomes, or difficulty assessing the outlook.
- Acknowledged, intrinsic, methodological or routine forecast uncertainty is insufficient. Code 1 only when unpredictability, unusually wide outcomes, exceptional uncertainty, or uncertainty surrounding the economic outlook is itself emphasized, or materially shapes the economic or policy argument being made.

General rules:

- * Base every classification only on the supplied text.
- * Classify substantive economic discussion rather than keyword presence.
- * Do not infer views that are not supported by the text.
- * For outlook variables, classify the dominant expected direction rather than counting positive and negative words.
- * Distinguish current conditions from the forward-looking outlook. For example, weak current activity can coexist with a positive growth outlook if the text clearly emphasizes an expected recovery.
- * When forward-looking evidence is genuinely mixed or insufficient, use 0.
- * Code an attention variable 1 only when the construct is an object of developed analysis: several connected claims, a focused passage, or a material role in the argument. An isolated consequence, a mandate reference, an illustrative example or a passing mention is 0. In a very short speech, proportional centrality can satisfy the threshold.
- * Attention measures whether a construct is substantively analysed, not whether the speech offers a forecast about it. A developed conceptual, historical or simulation-based treatment receives attention 1 even when the associated outlook is 0.
- * A hypothetical scenario, a causal channel, the intended effect of a structural reform, a policy simulation or a backward-looking episode does not by itself establish an outlook direction. Assign +1 or -1 only when the speaker presents that direction as the expected or baseline forward trajectory, or as the clearly dominant forward risk assessment.
- * Do not summarize the speech.
- * Do not add variables, explanations, commentary, confidence scores, or additional text.
- * Return only the required structured output.

Required output structure:

```
{
  "inflation_attention": 0,
  "inflation_outlook": 0,
  "growth_attention": 0,
  "growth_outlook": 0,
  "financial_stability_attention": 0,
  "uncertainty_attention": 0
}
```

Speech metadata and text follow below.

TITLE:
{title}

SUBTITLE:
{subtitle}

TEXT:
{text}

G Codebook Revision

Table 18 distinguishes two types of revision. Three repairs reconcile the human codebook used in rounds 1–3 with rules that were already present in the production LLM prompt. These repairs therefore change the human specification rather than the production prompt. Six additional decision rules were then developed from ambiguities identified during adjudication and added to both the codebook and the proposed prompt.

The changes remain limited in scope. They do not redefine the six underlying variables or alter the logical constraints linking attention and outlook. The production prompt already distinguished current conditions from the forward-looking outlook as a general rule; the human codebook expressed the same distinction within the variable-specific rules for inflation and growth outlook. The revised version instead clarify decision boundaries that the validation exercise showed could be applied inconsistently.

Table 18: Changes from the codebook used in rounds 1–3 to version 4

Section	Type	Change
§5	Repair	Removed the exclusion for technical, operational, legal, credit, settlement and financial-system risks, which belongs to §6 and contradicted §5’s own inclusion list.
§5	Repair	Added the override the prompt has carried since version 2 and the codebook never had: markets, rates and transmission are excluded <i>unless systemic or stability-related concerns are substantively addressed</i> .
§6	Repair	Regained the clause displaced into §5, so operational and credit risk are again excluded from uncertainty attention.
§2	Addition	<i>Conflicting horizons</i> . Where near-term headline inflation and a stated medium-term outlook point in opposite directions, classify on the medium-term policy-relevant horizon.
§5	Addition	<i>Payments and operations</i> . Settlement, payment continuity and operational resilience are not financial-stability attention unless tied to system-wide impairment, contagion, banking-system resilience, macroprudential risk, or the capacity of intermediation to serve the economy.
§6	Addition	<i>Uncertainty threshold</i> . Acknowledged, intrinsic, methodological or routine forecast uncertainty is insufficient. Code 1 only where unpredictability or exceptional uncertainty is itself emphasised, or materially shapes the argument.
General §8	Addition	<i>Substantive-attention threshold</i> . Code attention 1 only where the construct is an object of developed analysis. An isolated consequence, mandate reference, example or passing mention is 0; in a short speech proportional centrality suffices.
General §9	Addition	<i>Attention without forecast</i> . A conceptual, historical or simulation-based treatment receives attention 1 even where the outlook is 0. The constraint binds only in reverse.
General §10	Addition	<i>Outlook versus mechanism</i> . A hypothetical scenario, causal channel, intended reform effect or backward-looking episode does not establish direction. Assign ± 1 only where presented as the expected baseline or dominant forward risk.